ATAC-seq Analysis of Bulk B Cells and Plasmablasts

Chromatin accessibility across B-cell differentiation

Internship Project Report
Project P04

Student	sysbiol19
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Server	davros (Debian 13, kernel 6.12.74)
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1. Introduction

1.1 What the assay measures

ATAC-seq measures which parts of the genome are physically open. A Tn5 transposase pre-loaded with sequencing adapters is added to permeabilised nuclei. Wherever the DNA is accessible, the enzyme inserts its adapters and cuts the DNA at the same time. Where chromatin is wrapped around nucleosomes or packed into heterochromatin, the enzyme cannot get in.

This means the measurement is read density. Regions that collect far more fragments than expected by chance were open in the original cells. Everything I did in Week 1 exists to make sure that read density reflects real biology rather than technical noise (each filtering step removes one specific class of artefact that would otherwise look like a genuine signal).

Two properties of ATAC-seq shaped my parameter choices, and both come back repeatedly:

Fragment lengths vary a lot. One tagmentation event can produce a fragment covering just an open gap between nucleosomes (under 100 bp), a gap plus one nucleosome (around 200 bp), or several nucleosomes (400 bp and beyond). Fragments up to roughly 2 kb are real. Any tool that quietly discards long fragments throws away actual data.

There is no input control. ChIP-seq has a matched input sample to compare against. ATAC-seq has nothing equivalent, so the peak caller must estimate background from the sample itself.

1.2 The dataset

The data come from Calderon et al. (2019), Landscape of stimulation-responsive chromatin across diverse human immune cells, Nature Genetics. That study profiled chromatin accessibility across a large panel of sorted human immune cell populations, resting and stimulated, to map the regulatory landscape behind immune cell identity.

My assigned comparison is **bulk B cells against plasmablasts** (two stages of the same lineage). Bulk B cells are a resting, mixed population. Plasmablasts are what B cells become after activation: proliferating cells whose main job is secreting antibody. Getting from one to the other means shutting down the B-cell identity programme and switching on the plasma-cell programme, so I expected substantial chromatin remodelling, particularly around PRDM1, XBP1 and IRF4 on the plasmablast side and PAX5 and BACH2 on the B-cell side.

Week 1 used a single library, **SRR7650763**, subsetted to chromosome 22.

It is worth being explicit about what that library actually is, because it is easy to assume it belongs to the project. It does not. Checking the SRA run table shows SRR7650763 is annotated `cell_type = Immature_NK`, `lineage = NK` (an immature natural killer cell from donor 1001, not a B-lineage sample at all). This makes sense once the purpose of Week 1 is clear: the exercise was to learn what each pre-processing step does by running it manually, and for that the cell type is irrelevant. Any ATAC-seq library would have served. The tutorial materials for Week 2 Days 3–5 use an NK cell example throughout, so SRR7650763 evidently comes from that teaching dataset.

The consequence for reading this report is a clean division. **Week 1 is a methods exercise on an NK library. Weeks 2 and 3 are the actual project**, comparing bulk B cells against plasmablasts across seven B-lineage samples, using pre-generated nf-core output so that all samples were processed identically. The comparison in Section 3 between my manual pipeline and the reference is therefore a technical validation (same library, two pipelines) and not a biological result.

1.3 What the three weeks were for

Week 1 was not about getting a result quickly (nf-core/atacseq would have done the whole thing in one command inside Docker). The point was to run each pre-processing step by hand so that I would understand what each stage does and why each parameter is set the way it is, which is why the cell type of the practice library did not matter. Weeks 2 and 3 then built the consensus peak set, quantified accessibility across the seven project samples, tested for differential accessibility, and interpreted the result through target genes, motif enrichment and pathway analysis.

2. Week 1 - Manual Pre-processing

2.1 Environment and input

Every session started with:

```
source /pcpool/home/sysbiol_tutor/miniconda3/bin/activate
conda activate /pcpool/home/sysbiol_share/envs/atac
```

Tool	Version
Cutadapt	4.7
Bowtie2	2.5.5
SAMtools	1.24
Picard	3.4.0
BEDTools	2.31.1
MACS2	2.2.7.1 - **non-functional**, see §7.2
MACS3	3.0.2 (used instead)
MultiQC	1.35

Input property	Value
Accession	SRR7650763
Layout	Paired-end
Raw read pairs	1,708,421
Total bases	225,626,584 bp
Mean read length	~66 bp
Scope	Chromosome 22 only

2.2 Step 1 - Adapter trimming

Before trimming I had to work out what the adapters actually were rather than assume. ATAC-seq libraries are built by Tn5 carrying Nextera-type adapters (the chemistry is part of the assay, not a separate ligation step). The Tn5 mosaic-end sequence is `CTGTCTCTTATACACATCT`, and because tagmentation is symmetric the same sequence applies to both mates.

Why this matters: if a fragment is shorter than the read length, the sequencer runs off the end of the insert and starts reading adapter. Those bases are not genomic, so they either fail to align or align somewhere they do not belong.

```
cutadapt -a CTGTCTCTTATACACATCT -A CTGTCTCTTATACACATCT \
  -o trimmed_R1.fastq.gz -p trimmed_R2.fastq.gz \
  SRR7650763_1_chr22.raw.fastq.gz SRR7650763_2_chr22.raw.fastq.gz \
  > output_cutadapt/cutadapt.log
```

`-a` and `-A` give the adapter for R1 and R2. `-o` and `-p` keep the outputs synchronised, so a discarded pair is discarded from both files.

Metric	Value
Read pairs processed	1,708,421
R1 reads with adapter	1,893 (0.1%)
R2 reads with adapter	1,777 (0.1%)
Pairs written	1,708,421 (100.0%)
Bases trimmed	0.20%

With Cutadapt's default settings no pair fell below the length threshold, so every input pair was written out. This is confirmed downstream: the Bowtie2 alignment categories sum to exactly 1,708,421 pairs (§2.3), which is the number of pairs the aligner received.

An adapter rate of 0.1% is very low, which is good news. Read-through only happens when the insert is shorter than the read, so a rate this low means almost all fragments exceeded 66 bp (the library was well sized). Over-tagmented libraries routinely show 20–40% here.

A deviation to record. Calderon et al. ran Cutadapt with a minimum length of 20 bp and minimum adapter overlap of 5 bp; I used the defaults. To check whether it mattered I re-ran the trimming with the published settings, which discarded 7 read pairs out of 1.7 million as too short. That re-run was for comparison only: the trimmed FASTQ files that went into the alignment came from the default-settings run, so those 7 pairs never enter the analysis. The deviation is real but negligible either way.

2.3 Step 2 - Genome alignment

```
bowtie2 -p 2 -X 2000 -x ref/GRCh38_chr22 \
  -1 output_cutadapt/trimmed_R1.fastq.gz \
  -2 output_cutadapt/trimmed_R2.fastq.gz \
  -S sample.sam 2> SRR7650763_bowtie2.log
```

The only parameter changed from the defaults is `-X 2000`, the one ATAC-seq-specific setting at this stage. Bowtie2's default maximum insert size is 500 bp; anything longer is flagged as an improper pair, which the `-f 2` filter in the next step then discards. ATAC-seq routinely produces fragments spanning three or more nucleosomes, so the default would silently delete that population. Calderon et al. specify a 2 kb maximum, so that is what I used. Section 2.8 quantifies what this preserved in practice (8.8% of fragments, drawn specifically from the higher-order nucleosomal tail).

`-p 2` fixes the thread count for benchmarking. `2>` captures the alignment summary for MultiQC.

Result: 96.53% overall alignment rate. Comfortably above the 90% benchmark for human ATAC-seq, confirming the reference index was right, trimming did not damage the reads, and the library is not badly contaminated.

The breakdown (Figure 1, panel B1) shows where those alignments came from:

Category	Read pairs	%
Mapped uniquely	1,439,446	84.3%
Multimapped	166,873	9.8%
Neither mate aligned	59,209	3.5%
Discordantly uniquely	22,050	1.3%
One mate mapped uniquely	16,180	0.9%
One mate multimapped	4,662	0.3%

The important number here is not the headline 96.5% but the 84.3% mapping uniquely. A high overall rate can be inflated by repetitive multi-mapping reads, which the MAPQ ≥ 30 filter would then discard anyway. At 84.3% unique, most of the alignment rate is real.

A deviation on the genome build. The paper used hg19; I used **GRCh38**, confirmed from the BAM header (`@SQ SN:22 LN:50818468`). This was not a free choice - the supplied blacklist is in hg38 coordinates, and mixing builds between an alignment and a coordinate-based filter would mask entirely the wrong regions without any error message.

2.4 Hardware benchmarking

The task was to run the same alignment, pinned to exactly two threads, on the server and on a local PC and compare wall-clock time. Fixing the thread count is what makes the comparison meaningful - it removes parallelism as a variable, so any difference reflects per-core throughput, memory bandwidth and I/O rather than core count.

```
time bowtie2 -p 2 -X 2000 -x ref/GRCh38_chr22 \
  -1 trimmed_R1.fastq.gz -2 trimmed_R2.fastq.gz \
  -S /dev/null 2> /dev/null
```

Output was written to `/dev/null` so that disk write speed would not contaminate the measurement (the SAM file already existed from the main run). The reference index and trimmed FASTQs were copied to the local machine so both platforms processed byte-identical input, and both used the same Bowtie2 binary from the shared Conda environment.

Results:

Platform	real	user	sys	user/real
davros server (run 1)	5m04.9s	10m17.7s	1.3 s	2.03
davros server (run 2)	4m58.5s	10m04.9s	1.4 s	2.03
davros mean	**5m01.7s (301.7 s)**			
Local PC (dalek74)	**5m48.5s (348.5 s)**	11m47.1s	3.1 s	2.03
Difference	**+46.8 s - local 15.5% slower**			

Local hardware: Intel Core i3-3240 @ 3.40 GHz, 2 physical cores / 4 threads (a 2012 desktop part).

Interpreting the three numbers. `real` is elapsed wall-clock time and is the figure being compared. `user` is CPU time summed across all threads, which is why it is roughly double `real` here. `sys` is time spent in kernel operations, mostly I/O.

The **user/real ratio is 2.03 on every run**, including both platforms. With two threads that is essentially perfect parallel efficiency, and it matters for interpretation: it means the difference between machines is not a difference in how well they parallelised. Both saturated their two threads. What is left is per-core throughput.

The server is faster, but only by 15.5%, and that is the interesting part. I had expected a larger gap. Two things explain the modest difference.

First, constraining the run to two threads removes almost everything a server is good at. Servers are built for aggregate throughput (many cores, large memory, many concurrent jobs) not for single-task speed. Per-core clock speed on server CPUs is often lower than on consumer desktop parts, because server designs trade clock for core count and thermal headroom. Pinning both machines to two threads therefore compares the one dimension where the server has the least advantage.

Second, davros is shared. Other students' jobs were running during these measurements, so the server figure should be read as a realistic-load number rather than a best case. The 46.8 s gap is probably a lower bound on what an idle server would achieve.

The two independent server runs differed by only 6.4 s (2.1%), which gives some confidence that the 46.8 s gap between platforms is a real effect rather than measurement noise.

The `sys` time is worth a brief note: 3.1 s locally against 1.3 s on the server. Both are negligible relative to a 5-minute runtime, confirming this workload is CPU-bound rather than I/O-bound, but the twofold difference is consistent with a faster storage subsystem on the server.

Practical conclusion. For a single two-thread alignment, this decade-old desktop is within 16% of a shared research server. The server's value is not that any individual job runs dramatically faster; it is that it can run many such jobs simultaneously, hold much larger genomes in memory, and be left running for hours without tying up a workstation. For the full-genome versions of these libraries (80 million read pairs rather than 1.7 million on one chromosome) the local machine would also have been constrained by memory and by having only two physical cores.

Bonus task (CPU scaling. The optional scaling experiment (varying `-p` across 1, 2, 4 and 8) was not completed). In this particular case its value would have been limited: the local CPU has only **two physical cores**, so `-p 4` and `-p 8` would have been served by hyperthreading rather than by additional physical execution units, and the expected result is a near-linear gain from 1 to 2 threads followed by a sharp flattening beyond that. The general principle at work is Amdahl's law - the serial portions of the program (loading the FM-index, writing output) do not accelerate with additional threads and progressively come to dominate total runtime.

2.5 Step 3 - Filtering and duplicate removal

Raw Bowtie2 output contains many alignments that should not be in a peak-calling input. Four passes remove them.

Quality and flag filtering does most of the work:

```
samtools view -b -q 30 -F 1804 -f 2 sample.sam > sample.filtered.bam
```

- `-q 30` keeps alignments with MAPQ ≥ 30, a Phred-scaled confidence corresponding to a 1-in-1000 chance the position is wrong. It also removes multi-mapping reads, which get low MAPQ by definition.
- `-f 2` keeps only proper pairs (both mates aligned in the expected orientation and distance).
- `-F 1804` excludes reads carrying any of five flag bits: 4 (unmapped) + 8 (mate unmapped) + 256 (secondary) + 512 (failed vendor QC) + 1024 (already flagged duplicate). The number is their sum.

This is the ENCODE standard for ATAC-seq and matches the paper exactly.

On mitochondrial reads. Calderon et al. also removed chrM reads. In whole-genome ATAC-seq this matters enormously (mitochondrial DNA has no chromatin, so Tn5 shreds it and 20–50% of reads can end up there). My input was chromosome 22 only, so there were none. I mention it so its absence is not read as an omission.

Sorting, read groups, deduplication:

```
samtools sort -@ 2 -o sample.sorted.bam sample.filtered.bam

picard AddOrReplaceReadGroups I=sample.sorted.bam O=sample.rg.bam \
  RGID=1 RGLB=lib1 RGPL=ILLUMINA RGPU=unit1 RGSM=SRR7650763

picard MarkDuplicates I=sample.rg.bam O=sample.dedup.bam \
  M=SRR7650763.dup_metrics.txt REMOVE_DUPLICATES=true
```

```
samtools index sample.dedup.bam
```

Sorting comes first because Picard compares coordinates of neighbouring alignments and refuses unsorted input. The `AddOrReplaceReadGroups` step was not in the original protocol (I added it after MarkDuplicates crashed (§7.1)).

The logic of deduplication: PCR during library prep makes multiple identical copies of the same original molecule. Those are not independent observations of chromatin state, they are photocopies. Leaving them in inflates depth wherever amplification was efficient and produces peaks that are not real. Picard finds alignment groups with identical start and end coordinates, keeps the one with the best base qualities, and with `REMOVE_DUPLICATES=true` deletes the rest.

Metric	Value
Alignment records examined	2,956,782
Read pairs examined	1,478,391
Records marked duplicate	2,264,694
PERCENT_DUPLICATION	**76.59%**
Optical duplicate clusters	0
Reads remaining	692,088 (= 346,044 pairs)

The arithmetic checks out: 1,478,391 × 2 = 2,956,782, and 2,956,782 – 2,264,694 = 692,088, matching `samtools view -c`.

A duplication rate of 76.59% is the number that defines this dataset. Roughly three of every four sequenced fragments are redundant. Zero optical duplicate clusters means the redundancy is PCR-derived rather than an optical artefact of the flow cell. Section 8.1 uses the reference data to test two competing explanations for it.

2.6 Blacklist filtering

```
bedtools intersect -v -abam sample.dedup.bam \
  -b hg38-blacklist.ensembl.bed > sample.clean.bam
samtools index sample.clean.bam
```

The ENCODE blacklist lists regions that produce absurdly high signal in essentially every chromatin experiment regardless of cell type (repetitive elements, satellite and centromeric sequence, unresolved assembly). Signal there is an artefact of the reference genome and produces convincing-looking false peaks. `-v` inverts the normal intersect behaviour so only non-overlapping reads are kept.

Before running I checked that the coordinate systems matched, because a coordinate-based filter across genome builds fails silently:

```
samtools view -H sample.dedup.bam | grep '^@SQ'    # @SQ SN:22 LN:50818468 $\rightarrow$ hg38
head -3 hg38-blacklist.ensembl.bed                 # Ensembl naming, no 'chr' prefix
```

Result: 692,088 → 690,686 reads, 1,402 removed (0.20%). A small fraction, as expected for chromosome 22, but those reads are concentrated in a handful of positions and would have produced high-scoring artefactual peaks.

2.7 Step 4 - Peak calling

Everything to this point was cleaning. This is the first step producing an actual biological result.

```
macs3 callpeak -t sample.clean.bam -f BAMPE -g 4.0e7 \
  -n SRR7650763 --broad --broad-cutoff 0.1 --keep-dup all \
  --outdir macs2_out 2> macs2.log
```

- `-t` **with no** `-c` (no input control exists (§1.1)).
- `-f BAMPE` (reads real fragment coordinates from mate pairs instead of extending single reads by a modelled length). Because true fragment lengths are known, the model-building step is skipped and `--nomodel`, `--shift` and `--extsize` are unnecessary.
- `-g 4.0e7` (effective genome size, a deliberate departure from `-g hs` (2.7×10^9)). Giving the whole-genome value for a chromosome-22-only dataset would spread the background expectation across a genome that is 99% absent from the input, deflating background and inflating significance everywhere. Chromosome 22 is 50.8 Mb but much of its short arm is unmappable heterochromatin, so I used approximately 40 Mb.
- `--broad --broad-cutoff 0.1` (links adjacent weakly enriched regions into domains rather than reporting only sharp summits). Chosen for consistency with the reference output, which is supplied as `.broadPeak`.
- `--keep-dup all` (essential, not optional). Picard already removed duplicates; letting MACS apply its own filter on top would delete genuine independent fragments that happen to share coordinates.

Metric	Value
Peaks called	**1,035**
FRiP score	**0.574**
MACS estimated fragment length	202 bp
Peak width - min / median / max	203 / 759 / 4,571 bp

The FRiP score (Fraction of Reads in Peaks) was computed afterwards as a signal-to-noise measure:

```
IN=$(bedtools intersect -a sample.clean.bam -b macs2_out/SRR7650763_peaks.broadPeak -u | samtools view -c)
TOT=$(samtools view -c sample.clean.bam)
echo "scale=4; $IN/$TOT" | bc      # 396,476 / 690,686 = 0.5740
```

A FRiP above 0.3 is generally taken as indicating a high-quality ATAC-seq library, so 0.574 is well above threshold. It is also slightly higher than the 0.554 the nf-core reference obtained on the same library (see §3.2).

Example rows (chromosome, start, end, name, score, strand, fold-enrichment, $-\log_{10}$ p, $-\log_{10}$ q):

```
22   17084659   17085717   SRR7650763_peak_9    347   .   5.85381   37.1117   34.7239
22   16601106   16602418   SRR7650763_peak_4    119   .   5.73786   14.054    11.9367
22   10961379   10961838   SRR7650763_peak_1     26   .   3.47882    4.50638   2.62105
```

The confidence range is wide. Peak 9 combines 5.85-fold enrichment with q $\approx 10^{-35}$ (unambiguous). Peak 1 has 3.48-fold enrichment and q ≈ 0.0024, formally significant but near the detection limit and the kind of peak a shallow library loses. A median width of 759 bp confirms broad mode worked as intended; narrow-mode calls are usually 200–400 bp.

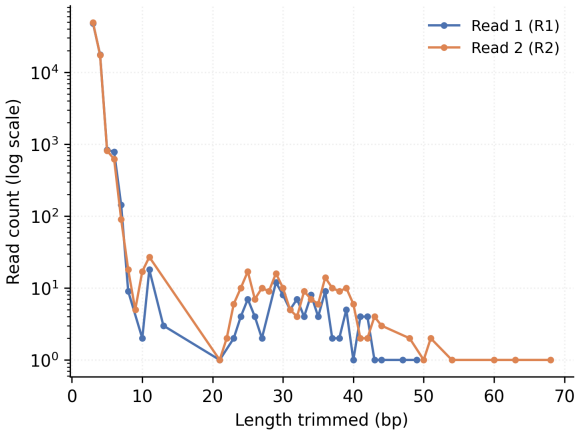
2.8 Step 5 - Quality control aggregation

Because the manual pipeline produces far fewer QC metrics than nf-core does, I also generated the insert-size distribution explicitly (it is the most informative single ATAC-seq quality readout).

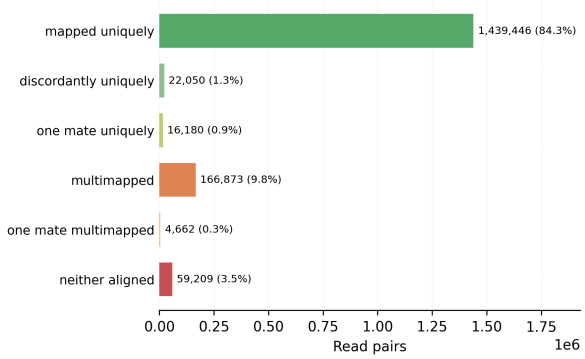
```
picard CollectInsertSizeMetrics I=sample.dedup.bam \
  O=SRR7650763.insert_size_metrics.txt \
  H=SRR7650763.insert_size_histogram.pdf M=0.5

multiqc . -o multiqc_out --force
```

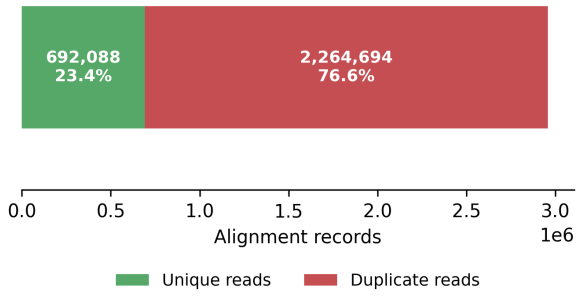
I ran it on the deduplicated BAM so the fragment lengths reflect distinct molecules rather than PCR copies. Four modules were detected: Cutadapt, Bowtie2, Picard and MACS.



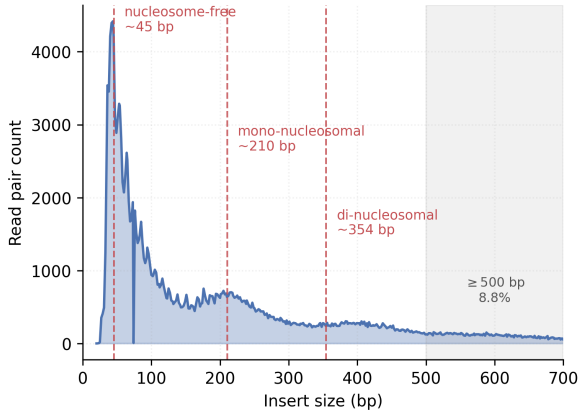
(a) Cutadapt (adapter trimming). 1,893 R1 and 1,777 R2 reads carried adapter (0.1%).



(b) Bowtie2 (paired-end alignment breakdown). Overall alignment rate 96.53%.



(c) Picard MarkDuplicates (unique versus duplicate read pairs (76).59% duplication, 0 optical).



(d) Picard CollectInsertSizeMetrics (insert-size distribution). Median 124 bp, mean 199.1 bp.

Figure 1. Week 1 library quality control for SRR7650763. All panels re-plotted from the underlying MultiQC v1.35 data. The shaded band in (d) marks fragments $\geq$ 500 bp, the 8.8% preserved by `-X 2000`.

Panel	Metric	Value
A1	Bases trimmed	0.20%
B1	Overall alignment rate	96.53%
C1	Duplication	76.59%
D1	Median insert size	124 bp
D1	Mean insert size	199.1 bp

Panel D1 is the most interesting result of Week 1. The distribution is clearly multi-modal, which is the signature of a working ATAC-seq library:

Feature	Value
Modal insert size	~44 bp
Fragments < 100 bp (nucleosome-free)	44.3%
Fragments 180–250 bp (mono-nucleosomal)	12.7%

Feature	Value
Fragments 350–450 bp (di-nucleosomal)	7.5%
Detected modal peaks	~45, ~210, ~354 bp

One caveat on the p-values before interpreting them: 1e-324 is the smallest number representable in double precision, so the top of the down-set list has hit the numerical floor. Ranking Elf4 above PU.1 on that basis is not meaningful; both are simply beyond the limit of the arithmetic.

Three things stand out. Nearly half of all fragments are shorter than 100 bp, coming from open stretches small enough for two Tn5 insertions with no nucleosome between them - this sub-nucleosomal fraction carries the actual accessibility signal, and a large one is the best single indicator of a clean library. There is a mono-nucleosomal shoulder around 210 bp, fragments spanning one nucleosome (roughly 147 bp of wrapped DNA plus linker). And a di-nucleosomal peak reappears around 354 bp.

A word of care about what this does and does not prove regarding `-X 2000`. The di-nucleosomal mode sits at ~354 bp, which is below Bowtie2's default 500 bp insert limit, so that peak would have survived the default setting. What `-X 2000` actually preserves is the long tail beyond 500 bp: **8.8% of fragments in this library are $\geq$ 500 bp**, and every one of them would have been classified as improperly paired and then removed by the `-f 2` filter. So the setting protects roughly one fragment in eleven, drawn specifically from the higher-order nucleosomal population, rather than rescuing the di-nucleosomal peak itself.

The distribution also shows that the single MACS estimate of 202 bp is a compromise between the sub- and mono-nucleosomal modes rather than a description of a mono-nucleosomal library.

2.9 Where the reads went

Stage	Read pairs	% of raw
Raw input	1,708,421	100.00%
After adapter trimming	1,708,421	100.00%
Aligned (96.53% rate)	~1,649,000	~96.5%
After MAPQ $\geq$ 30, proper-pair, flag filter	1,478,391	86.53%
After duplicate removal	**346,044**	**20.25%**
After blacklist filtering	345,343	20.21%
Peaks called	**1,035**	-

The cascade is unmistakable. Trimming, alignment and blacklist filtering together cost under 4%. Quality and flag filtering cost another 13%, which is normal. Duplicate removal alone cost 66 percentage points. If I wanted more data from this sample, no parameter tuning would help (the problem is upstream, at the bench).

3. Comparison with the Reference Pipeline

Partway through, a pre-generated MultiQC report became available covering the full nf-core/atacseq run over the entire study (97 libraries). It includes **SRR7650763 itself**, which turns a vague comparison into a direct same-sample test.

This section is a technical validation of the Week 1 pipeline, not a biological result (SRR7650763 is an NK library (§1.2), so nothing here bears on the bulk B versus plasmablast question). What it establishes is whether a manually assembled pipeline reproduces what an established workflow does on identical input.

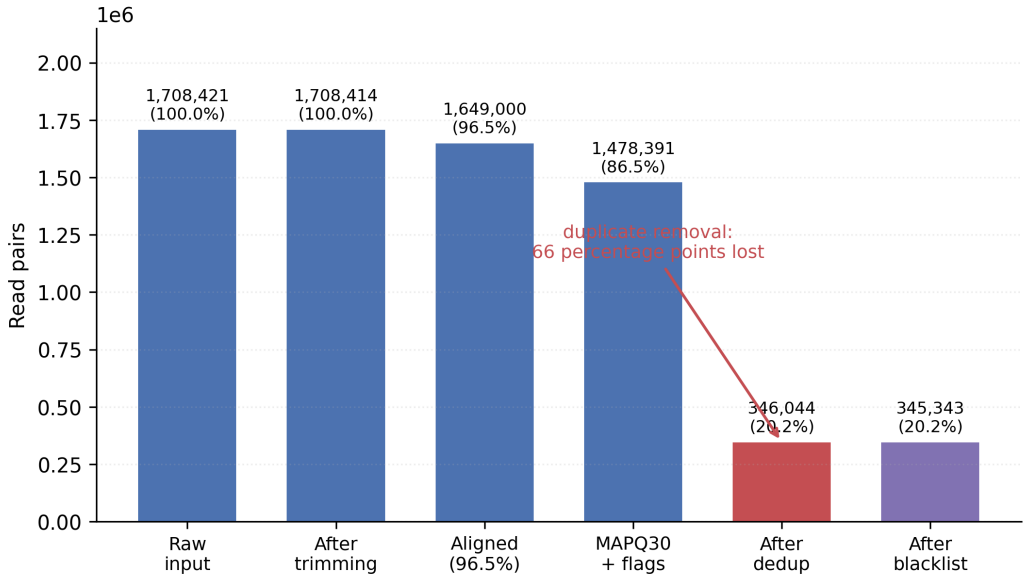


Figure 2. Read retention through the Week 1 pipeline. Duplicate removal alone accounts for 66 of the 80 percentage points lost.

3.1 What the reference did differently

Component	Reference (nf-core/atacseq 2.1.2)	Mine
Workflow engine	Nextflow 25.10.4	manual shell commands
Trimming	TrimGalore 0.6.7 (Cutadapt 5.2)	Cutadapt 4.7 directly
Aligner	**BWA-MEM 0.7.17**	**Bowtie2**
Duplicates	Picard 3.0.0	Picard 3.4.0
Peak calling	MACS2 2.2.7.1	MACS3 3.0.2
Extra QC	FastQC, deepTools, ataqv, HOMER	none

The aligner difference matters most. Both are solid, but they score alignments differently, so a MAPQ ≥ 30 filter does not remove exactly the same reads in the two pipelines. A genuine methodological difference rather than an error on either side, but it limits how precisely the peak sets can be compared.

3.2 Same-sample comparison

Metric	Mine (manual, chr22)	Reference (nf-core, genome)
Raw read pairs	1,708,421	80,018,786
Alignment rate	96.53%	98.10%
Duplication	**76.59%**	**73.76%**
Read pairs after duplicate removal	346,044 (20.26%)	20,505,832 (25.63%)
Read pairs in the final filtered BAM	**345,343 (20.21%)**	**16,590,727 (20.73%)**
Peaks	1,035 (chr22)	52,082 (genome)
FRiP	**0.574**	0.554

Two rows are needed here rather than one, because the two pipelines apply their filters in a different order, so "the reads that survived" means different things at intermediate stages.

In my pipeline the MAPQ ≥ 30 and proper-pair filters run before duplicate removal, so the 346,044 pairs left after Picard are already fully filtered; blacklist removal then takes a further 0.2%. The nf-core pipeline removes duplicates earlier and applies more of its filtering afterwards, so its 20,505,832 post-duplicate pairs still contain reads that later steps discard. Comparing those two intermediate numbers directly (20.26% against 25.63%) would be comparing different things.

The comparable point is the end of the pipeline, and there the agreement is striking: 20.21% against 20.73%, half a percentage point apart. Duplication also lands within three points (76.59% against 73.76%). Two pipelines differing in aligner, trimming wrapper, peak-caller version and genomic scope arrive at essentially the same usable fraction of the input, which is strong evidence that the manual pipeline was built correctly.

The FRiP scores add to this: 0.574 for my pipeline against the reference's 0.554 on the same library.

It is worth being precise about what depth this was achieved at, because the raw totals invite a misleading comparison. My input was 1.7 million read pairs restricted to chromosome 22; the reference processed 80 million pairs genome-wide. That is only 2.1% of the reference's total, but the two are not measuring the same territory. Scaling the reference's 16.6 million final filtered pairs by chromosome 22's share of the genome (1.5–2.2%, depending on whether one uses mappable sequence or observed peak share) gives roughly 250,000–370,000 pairs on that chromosome. My 345,343 sits inside that range.

So the honest statement is that the per-chromosome depths are broadly comparable, which is exactly why the FRiP scores land so close together. The manual pipeline is not extracting better signal from less data; it is extracting comparable signal from comparable data, which is the more useful conclusion for validating the pipeline.

3.3 Peak count, done properly

Comparing 1,035 against 52,082 is meaningless because the second is genome-wide. I measured the chromosome-22 share directly from a reference sample where the per-chromosome file was available:

```
grep -c "^22    " ${REF}/SRR7650729_REP1.mLb.clN_peaks.broadPeak   # 1,354
```

SRR7650729 has 1,354 of 61,353 peaks on chromosome 22 (a share of 2.21%).

Quantity	Value
Reference peaks for SRR7650763 (genome-wide)	52,082
Chromosome-22 share (measured)	2.21%
Expected on chr22	**~1,151**
I found	**1,035**
Recovery	**89.9%**

Recovering about 90% with a different aligner and a different major version of the peak caller is a good outcome - but the calculation rests on an assumption that needs stating plainly, because it is weaker than it first appears.

The chromosome-22 share of 2.21% was measured in **SRR7650729, a bulk B cell**, and then applied to **SRR7650763, an immature NK cell** (§1.2). These are different lineages, and there is no guarantee that chromosome 22 carries the same proportion of accessible regions in both. Chromosome 22 is gene-dense and immunoglobulin-lambda-bearing, so if anything a B-lineage sample might be expected to have a higher share there than an NK sample (which would mean 1,151 overestimates the expectation and the true recovery is better than 89.9%).

This should therefore be read as a **rough cross-lineage approximation**, not a precise figure. The reference report gives only the genome-wide peak count for SRR7650763, not a per-chromosome breakdown, so a

direct like-for-like number was not available to me. What the calculation supports is the weaker but still useful claim that my peak count is of the right order of magnitude (not that it is 89.9% of a well-defined target).

3.4 Where my sample sits in the project

Sample	Peaks	FRiP	Duplication	Unique pairs
SRR7650851	72,208	12.6%	13.1%	32,058,762
SRR7650735	64,045	25.0%	26.1%	48,370,602
SRR7650729	61,353	35.6%	31.2%	39,655,365
SRR7650918	59,262	10.7%	22.1%	61,216,470
SRR7650763 (NK)	**52,082**	**55.4%**	**73.8%**	**20,505,832**
SRR7650812	51,607	20.4%	23.1%	21,373,245
SRR7650777	47,573	23.9%	51.4%	26,507,273
SRR7650771	27,330	4.4%	19.6%	38,876,406

The Week 1 library has the worst duplication of any of these and, simultaneously, by far the best FRiP. That looked contradictory at first; §8.2 works through why it is not.

4. Week 2, Days 1–2 - Consensus Peaks and Feature Counts

Week 2 moves from one sample to the group. From here I stopped using my own manual output and worked with the pre-generated BAMs and peak files, because comparing samples requires that all of them were processed identically.

4.1 Samples and metadata

Seven samples are in the project directory. SRR7650763 (the library I processed manually in Week 1) is not among them, and as explained in §1.2 the reason is that it is an NK cell library rather than a B-lineage one. The Week 1 and Week 2 datasets are separate: one was a methods exercise, the other is the project.

Rather than guess which sample was which cell type (an error there would invalidate everything downstream) I built the metadata table from the official **SraRunTable.csv** downloaded from the study's SRA entry. The table has 175 runs covering the whole study, so the task was to cut it down to our seven samples and the columns that matter.

One practical detail: the CSV contains quoted fields with embedded commas (`"run.zq,fastq,sra"`), so parsing it with `awk -F,` would break. I used R's `read.csv`, which handles quoting properly. The full script is in Appendix A.1.

Sample	Donor	Cell type	Lineage	Treatment
SRR7650729	D1001	Bulk_B	B	no_treament
SRR7650735	D1001	Plasmablast	B	no_treament
SRR7650771	D1002	Bulk_B	B	no_treament
SRR7650777	D1002	Plasmablast	B	no_treament
SRR7650812	D1003	Bulk_B	B	no_treament
SRR7650851	D1004	Bulk_B	B	no_treament
SRR7650918	D1010	Plasmablast	B	no_treament

Two things worth noting from this table. All seven are untreated, so stimulation is not a confounder - the full SraRunTable also contains `treatment1` versions of three of these samples (SRR7650730, 772,

813), and they were correctly excluded. Lineage is constant (B) across all samples, so it cannot serve as a variable here, but I kept it in the table because the assignment asks for it.

The important structural finding is the donor layout:

Donor	Bulk_B	Plasmablast
D1001	1	1
D1002	1	1
D1003	1	0
D1004	1	0
D1010	0	1

Only donors 1001 and 1002 have both cell types. The groups are also unbalanced, four against three. I checked whether more samples could complete the pairing and they cannot (among untreated B-lineage samples in the whole study, these are all that exist). So this is a limitation of the source dataset, not of the sample selection. It constrains the statistical design, and §5.3 deals with the consequences.

As a cross-check I had earlier pulled the same metadata independently from the ENA REST API, and the two sources agree exactly.

4.2 Building the consensus peak set

Each sample has its own peak file and the peaks do not line up between samples. To compare accessibility you first need a single shared set of regions every sample can be measured against: take the coordinates from every sample, pool them, sort them, merge anything that overlaps.

```
for sample in "${SAMPLES[@]}"; do
    awk '{print $1"\t"$2"\t"$3}' "${PEAK_DIR}/${sample}_REP1.mLb.clN_peaks.broadPeak"
done | sort -k1,1 -k2,2n | bedtools merge -i - > consensus_peaks.bed
```

A point I initially misunderstood: peaks found in only one sample are **not** discarded. They stay in the consensus set. Samples with no peak there simply get a near-zero count during quantification, and that near-zero is itself informative (it is exactly the kind of region that may turn out to be differentially accessible).

Result: 155,948 consensus peaks from seven samples whose individual peak counts ranged from 27,330 to 72,208.

featureCounts needs SAF format rather than BED, and each region needs a unique ID:

```
# NOTE: BED is 0-based half-open, SAF is 1-based inclusive -> start should be $2+1.
# Reproduced here as actually run; see section 4.2 for the impact assessment.
awk 'BEGIN{OFS="\t"; print "GeneID","Chr","Start","End","Strand"}
     {print "peak_"NR,$1,$2,$3,"."}' consensus_peaks.bed > consensus_peaks.saf
```

A one-base error I found afterwards. BED coordinates are 0-based and half-open; SAF coordinates are 1-based and inclusive at both ends. The conversion above passes the BED start through unchanged, so every interval in the SAF file begins one base earlier than it should. The correct form is `$2+1`:

```
{print "peak_"NR,$1,$2+1,$3,"."}
```

I have left the script as it was actually run rather than silently correcting it. The practical effect is one extra base at the left edge of intervals whose median width is 759 bp (roughly 0.13% of each counting window, applied identically to every peak and every sample). Because the offset is systematic rather than differential, it cannot bias a between-group comparison: any fragment gained or lost at that edge is gained or lost equally in bulk B and plasmablast samples. The DAR results are therefore unaffected in practice,

but the conversion is wrong as written and would matter for narrow features such as transcription factor footprints.

4.3 Counting fragments

```
featureCounts -F SAF -a consensus_peaks.saf -o counts_matrix.txt \
  -p --countReadPairs -B -C -T 4 "${BAMS[@]}"
```

- `-p --countReadPairs` counts fragments rather than individual reads, because the fragment is the unit of observation in ATAC-seq.
- `-B` counts only fragments whose **both ends aligned successfully**. (Note: this is a property of the alignment, not of the feature — `-B` does not require the two mates to fall inside the same peak. Since the input BAMs had already been filtered to proper pairs, the flag removed essentially nothing here.)
- `-C` discards chimeric pairs whose ends map to different chromosomes.
- `-T 4` uses four threads.

Sample	Total alignments	Assigned	Rate
SRR7650729	33,723,338	13,707,454	40.6%
SRR7650735	40,842,813	12,177,766	29.8%
SRR7650771	33,126,751	3,505,385	10.6%
SRR7650777	21,361,447	6,250,075	29.3%
SRR7650812	15,603,092	4,288,567	27.5%
SRR7650851	26,949,798	5,000,183	18.6%
SRR7650918	52,529,464	8,335,282	15.9%

Rates between 10% and 41% look low at first but are normal for ATAC-seq (most reads land in background rather than inside called peaks). What matters is that every `Unassigned` category except `NoFeatures` is zero: no unmapped reads, no mapping-quality failures, no chimeras. The input BAMs were clean.

The rates also track the FRiP scores from §3.4 closely (SRR7650771 is lowest on both, 10.6% and 4.4%), which is a nice internal consistency check.

Finally I trimmed the matrix to what DESeq2 needs (peak ID plus seven count columns with readable headers replacing the long BAM paths).**Result: 155,948 peaks × 7 samples.**

4.4 A first look at the biology

Before trusting the metadata I wanted the data to confirm it independently. PRDM1 (BLIMP1) is the master regulator of plasma-cell differentiation, so its locus should be more open in plasmablasts. Reading the columns in order (729 (B), 735 (P), 771 (B), 777 (P), 812 (B), 851 (B), 918 (P)):

Peak	729 B	735 P	771 B	777 P	812 B	851 B	918 P
peak_123060	6	**86**	14	**58**	3	8	**69**
peak_123062	9	**132**	8	**76**	4	3	**74**
peak_123064	17	**129**	21	**78**	9	13	**96**
peak_123065	13	**56**	14	**63**	9	17	**84**

Seven- to twenty-fold higher in every plasmablast sample. The group assignment is confirmed by the biology, not just by the metadata. This took five minutes and would have caught a catastrophic labelling error. Section 5.6 revisits these exact peaks with statistics attached.

I also looked at PAX5 on chromosome 9 expecting the opposite direction and got a mixed picture (some peaks higher in bulk B, some in plasmablasts). That confused me until I realised my window covered the whole PAX5 gene body, which contains many regulatory elements that need not move in the same direction. It is a useful preview of the "intrinsic enhancers versus promoters" problem flagged for Week 3.

5. Week 2, Days 3–5 - Differential Accessibility

5.1 Building the SummarizedExperiment

The count matrix and the metadata table are two separate files, and nothing enforces that their sample orders agree. A `SummarizedExperiment` binds them into one object so that any subsetting keeps counts and metadata synchronised. Before building it I assert the orders match:

```
stopifnot(all(colnames(counts) == rownames(coldata)))
```

If this ever failed, the entire analysis would silently compare the wrong groups. It is the single most important line in the script, and it passed.

I set `Bulk_B` as the reference level of `cell_type`, which makes a positive $\log_2$ fold change mean "more open in plasmablasts" (the direction that reads naturally).

5.2 Low-count filtering

Raw consensus matrices contain many regions supported by only one sample. They carry no statistical power and inflate the multiple-testing burden, so they are removed.

Here I hit a discrepancy between the two sets of instructions. The assignment PDF says to keep peaks with ≥ 10 counts in at least N samples where **N is half the total** (for seven samples, N = 4). The Day 3–5 slides say **at least 3 samples**, with the rationale that "since our smallest experimental group has 3 replicates, a real peak should be visible in at least one entire group."

Our smallest group is 3 plasmablasts, so the slide rule gives N = 3. I went with the slide's rationale because it is the better argument - a peak genuinely present in one whole cell type should be kept even if absent from the other, which is precisely the kind of region a cell-type comparison is looking for. I wrote it so the threshold is computed from the data rather than hard-coded:

```
N <- min(table(coldata$cell_type))
keep <- rowSums(assay(se, "counts") >= 10) >= N
```

Rule	N	Peaks retained	Removed
Assignment PDF (half of total)	4	83,434	46.5%
Slides (smallest group) - used	**3**	**101,593**	**34.9%**

I ran both. As shown in §5.4, the PCA result is essentially identical either way (PC1 = 88% at N=4, 87% at N=3), so the choice does not drive the conclusions (which is worth knowing if the question comes up).

5.3 Experimental design

The slides present `~ Donor + cell_type` as the design, with donor as a blocking factor to remove donor-to-donor noise. That is the right approach in general, and in the slides' own example it works because three donors each contributed both cell types.

It does not transfer cleanly to my data. I checked whether `~ Donor + cell_type` is even estimable on all seven samples, and it is (the design matrix is full rank, six parameters from seven samples). But it is a bad idea, because three donors contributed only one sample each. When a donor has a single sample,

its donor coefficient absorbs that sample's deviation entirely and the sample contributes nothing to the cell-type comparison. The model would spend three degrees of freedom to learn nothing, leaving one residual degree of freedom.

Design	Samples	Parameters	Residual df
`~ cell_type`	7	2	**5**
`~ Donor + cell_type`	7	6	1
`~ Donor + cell_type` (matched only)	4	3	1

The second and third rows carry the same information, but the four-sample version is honest about what it is doing. So:

- **Main analysis:** `~ cell_type` on all seven samples.
- **Sensitivity analysis:** `~ Donor + cell_type` on the four matched samples from donors 1001 and 1002.

The PCA supports this (donor variance is small next to the cell-type effect) and §5.7 tests it directly rather than just asserting it.

5.4 Normalisation, variance stabilisation and exploratory QC

DESeq2 estimates size factors to correct for library depth. They ranged from 0.550 to 1.745 with no extreme outliers, so depth differences are well within what the normalisation handles.

Variance stabilisation was applied with `blind = FALSE`, which lets the transformation account for the experimental design rather than treating all samples as exchangeable.

Panels A2 and B2 - the transformation worked. Before VST the red line, the running mean of standard deviation, slopes from about 1.0 on the left down to 0.6 on the right: low-count peaks carry more variance than high-count ones. After VST it is flat at roughly 0.5 across the whole range. That flattening is the entire point - without it, PCA and clustering would be dominated by whichever peaks happen to have the most counts rather than by biology.

Panel C2 - PC1 explains 87% of the variance. All four bulk B samples sit on the left, all three plasmablasts on the right, cleanly separated along PC1. PC2 accounts for 6%. Donor symbols are scattered across both groups with no visible pattern, so donor identity is not driving the separation (cell type is). This is what justified the design choice in §5.3.

Panel D2 - the clustering agrees. The dendrogram splits the samples into exactly two branches, three plasmablasts and four bulk B cells, and within-group distances are visibly smaller than between-group distances. Same conclusion as the PCA by a different method.

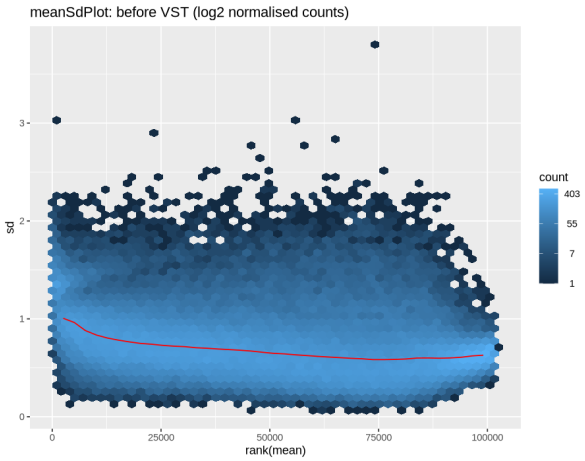
One observation: among the plasmablasts, donors 1001 and 1002 are closer to each other than either is to donor 1010 (the one donor with no matching bulk B sample). Worth remembering, but it does not disturb the separation.

5.5 Identifying differentially accessible regions

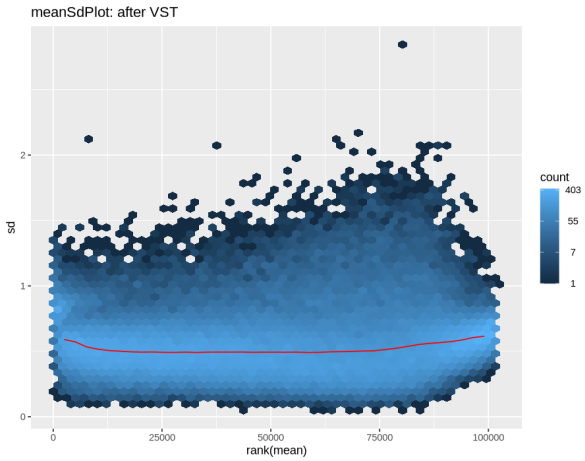
The contrast compares plasmablasts against bulk B cells as reference:

```
res  <- results(ddsA, contrast = c("cell_type", "Plasmablast", "Bulk_B"))
resS <- lfcShrink(ddsA, coef = "cell_type_Plasmablast_vs_Bulk_B", type = "apeglm")
```

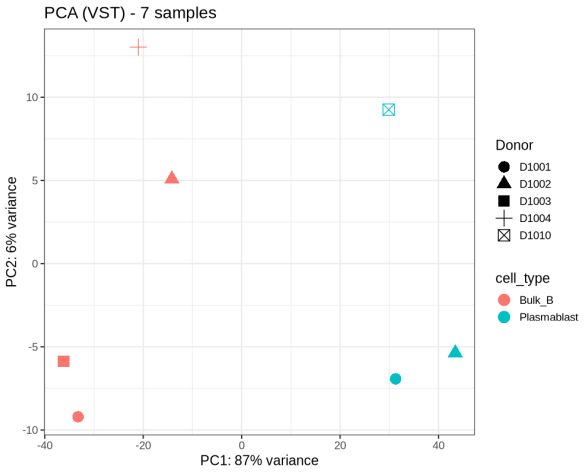
Log_2 fold changes were shrunk with **apeglm**. Low-coverage peaks produce wildly exaggerated fold changes purely from sampling noise; apeglm pulls those toward zero while leaving well-supported peaks alone, giving estimates that are conservative and biologically realistic.



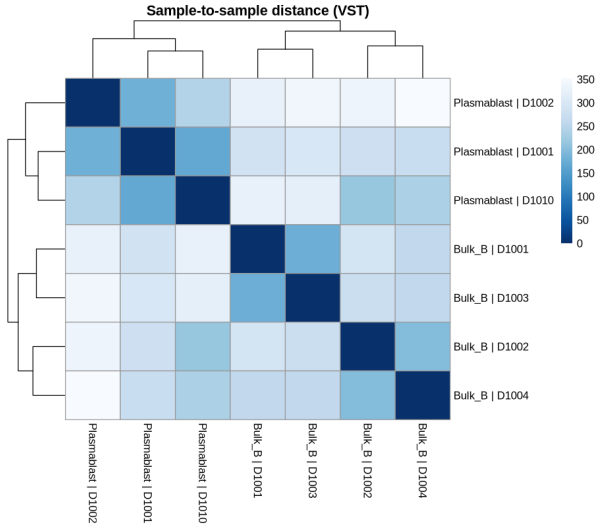
(a) Before VST, on $\log_2$ normalised counts.



(b) After VST (the running mean flattens).



(c) PCA of VST values; colour = cell type, symbol = donor.



(d) Sample-to-sample Euclidean distances with clustering.

Figure 3. Exploratory quality control of the seven-sample count matrix (101,593 peaks after filtering). PC1 accounts for 87% of the variance and separates cell types; donor symbols scatter across both groups.

DARs were then defined by the assignment's criteria: **FDR < 0.05 and** $|\log_2 \mathbf{FC}| \geq 1$.

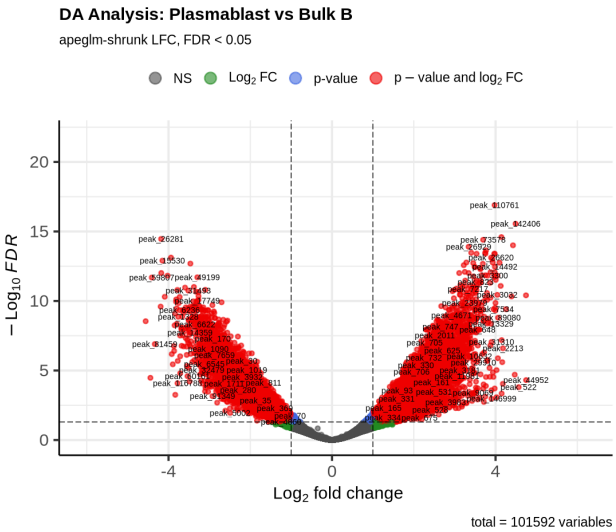
Metric	Value
Peaks retained after filtering	101,593
Peaks tested (non-NA adjusted p-value)	**101,592**
Total DARs	**14,662 (14.4%)**
More open in plasmablasts	7,976 (54.4%)
More open in bulk B	6,686 (45.6%)

One peak of the 101,593 retained by filtering has no adjusted p-value: DESeq2 flagged it as a count outlier by Cook's distance and set `padj = NA`. The tested total is therefore 101,592, and that is the denominator used throughout this section and in Figure 4.

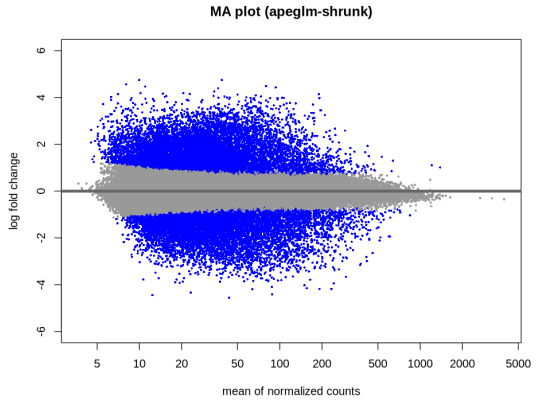
The strongest ten DARs all have $|\log_2 \mathrm{FC}|$ between 3.3 and 4.5 with adjusted p-values from 10^{-14} to 10^{-17} (ten- to twenty-two-fold differences in accessibility):

Peak	baseMean	$\log_2$FC	padj
peak_110761	191.2	3.98	1.3×10^{-17}
peak_142406	79.9	4.49	2.8×10^{-16}
peak_119208	190.3	4.14	2.5×10^{-15}
peak_26281	233.5	–4.18	3.4×10^{-15}
peak_73578	172.1	3.70	4.0×10^{-15}

A note on numbers that could be confused: DESeq2's own `summary()` reported 9,957 up and 8,373 down, but those use its defaults (padj < 0.1, no fold-change threshold). The 14,662 figure uses the assignment's stricter criteria applied to the shrunk estimates.



(a) Volcano plot. Dashed lines mark the DAR thresholds ($|\log_2 \mathrm{FC}| = 1$, FDR $= 0.05$); red points are significant.



(b) MA plot of shrunk $\log_2$ fold change against mean normalised counts; blue points are significant.

Figure 4. Differential accessibility, plasmablasts versus bulk B cells. 14,662 of 101,592 tested regions are significant: 7,976 opening and 6,686 closing. Both estimates are apeglm-shrunk.

Panel A3 shows a symmetric V with dense red on both wings and no tilt toward either side. That symmetry matters: a strong asymmetry would suggest a technical bias in normalisation rather than biology. The labelled extremes reach $|\log_2 \mathrm{FC}|$ of 4–5 on both sides.

Panel B3 shows the grey band of non-significant peaks narrowing as counts increase, which is exactly the expected behaviour (more reads means a more precise estimate). It also shows apeglm doing its job: the extreme fold changes at low counts have been pulled in, while genuinely well-supported large changes survive at high counts on the right.

Finding 14.4% of tested regions differentially accessible is a large fraction, but it is consistent with everything upstream. PC1 explained 87% of the variance, and bulk B cells and plasmablasts really are separated by a major differentiation programme rather than a subtle perturbation.

5.6 Biological validation with PRDM1

Section 4.4 predicted from raw counts that the PRDM1 locus would be more open in plasmablasts. Now the statistics can test it. Of the ten consensus peaks in the locus:

Peak	baseMean	$\log_2$FC	padj	DAR?
peak_123058	102.2	0.11	0.900	-
peak_123059	248.3	–0.25	0.721	-
peak_123060	28.9	**2.32**	**< 0.001**	**yes**
peak_123061	137.2	0.28	0.683	-
peak_123062	34.8	**3.24**	**< 0.001**	**yes**
peak_123063	38.6	–0.66	0.188	-
peak_123064	43.1	**1.92**	**< 0.001**	**yes**
peak_123065	32.4	**1.50**	**0.004**	**yes**
peak_123066	29.3	**1.47**	**0.003**	**yes**
peak_123067	34.5	0.08	0.930	-

Five of the ten peaks are DARs, and **all five point in the plasmablast direction** (no contradictions). The other five are essentially flat, with $\log_2$ fold changes between –0.66 and +0.28.

This pattern is more informative than a uniformly significant locus would have been. The PRDM1 region is not simply "switched on" as a block. Specific regulatory elements inside it gain accessibility while neighbouring elements stay unchanged, which is what selective enhancer activation looks like. It also happens to illustrate the Week 3 warning about pooling every peak in a gene body: had I averaged all ten, the five flat peaks would have diluted a clear signal into a weak one.

5.7 Sensitivity analysis: does donor matter?

The main analysis ignores donor. The sensitivity analysis tests whether that was safe, by refitting the paired design `~ Donor + cell_type` on the four matched samples from donors 1001 and 1002 and asking whether it recovers the same regions.

Comparison	Value
Design A DARs (7 samples, `~ cell_type`)	14,662
Design B DARs (4 samples, `~ Donor + cell_type`)	3,696
DARs found by both	3,692
Design B DARs also found by A	**99.9%**
$\log_2$FC sign agreement on A's DARs	**100.0%** (n = 13,036)

Design B finds far fewer DARs, which is expected (four samples instead of seven, and one residual degree of freedom). But of the 3,696 it does find, **3,692 are also in design A**. Only four are unique to B. And across every one of design A's DARs that design B could also test, the direction of change agrees **100% of the time**.

This is a strong concordance. The concern with an unblocked design is that donor-to-donor variation might masquerade as a cell-type effect; if that were happening here, controlling for donor should have flipped or dissolved a meaningful fraction of the calls. It did not. The paired model, which explicitly removes donor variation, produces a near-strict subset of the same regions with the same directions, which suggests the main analysis is detecting the cell-type difference rather than donor artefacts.

Two caveats keep this from being proof. The test itself rests on only two donors, so it cannot detect a confounder shared by both. And a subset relationship is what one would expect from any loss of power, concordant or not (the informative part is the direction agreement, not the overlap count). What can fairly be said is that the main biological conclusions are **robust to donor adjustment as far as this dataset can test it**, and that the design compromise forced by the incomplete pairing (§4.1) cost statistical power without producing detectable distortion.

6. Week 3 - Target Genes, Motifs and Pathways

Week 2 ended with 14,662 differentially accessible regions, all expressed as peak IDs and coordinates. Week 3 turns that into biology: which genes are involved, which transcription factors drive the change, and which pathways are affected.

6.1 Assigning peaks to genes

Nothing in the DAR table says which gene a region belongs to, so the first task was annotation. I built a TSS reference from the Ensembl GRCh38 release 110 GTF, keeping only protein-coding genes (19,445 in total).

Strand matters here and it is easy to get wrong. A gene's transcription start site is at its start coordinate on the + strand but at its end coordinate on the `-` strand, so the extraction has to branch on strand:

```
tss = ($7=="+") ? $4 : $5
```

This is one of the pitfalls the assignment raises, and getting it backwards would systematically misplace every TSS on the minus strand (roughly half the genome).

I then used `bedtools closest` to find the nearest TSS for each DAR and record the distance. Two gene lists came out of this:

List	Criterion	Genes
Promoter-proximal	DAR within 5 kb of a TSS	731 up / 733 down
All nearest genes	no distance limit	4,227 up / 3,704 down

I used the 5 kb lists for GO enrichment. The unrestricted list is larger but far less trustworthy: a peak 200 kb from the nearest TSS may regulate that gene through looping, or may regulate something else entirely. This is exactly the "intrinsic enhancers versus promoters" problem flagged in the assignment, and Section 6.3 shows it is not hypothetical.

6.2 Selecting six target genes

Rather than pick genes from the literature and hope they appeared, I checked which canonical markers were actually in my promoter-proximal lists:

```
grep -wE "PRDM1|XBP1|IRF4|CD38|SDC1|MZB1|TNFRSF17" genes_up_promoter.txt
grep -wE "PAX5|BACH2|MS4A1|CD19|CR2|CIITA|EBF1" genes_down_promoter.txt
```

The plasmablast side returned CD38, IRF4, SDC1 and TNFRSF17. The bulk B side returned CD19, CIITA, MS4A1 and PAX5. Eight canonical markers, every one on the expected side. That result on its own is a strong validation of the whole pipeline (a labelling or directional error anywhere upstream would have scrambled it).

Notably, **PRDM1 did not appear in either promoter-proximal list**. Its DARs are all more than 90 kb from the TSS, so the 5 kb filter excluded it (a fact that turns out to be the most interesting thing about the locus (§6.3)).

I therefore selected **five targets from the promoter-supported markers above** (IRF4, SDC1, PAX5, MS4A1, CIITA) and **retained PRDM1 a priori** despite its absence from those lists, because it is the master regulator of the transition under study and because its exclusion by a promoter-distance filter is itself a finding worth examining. The six:

Gene	Direction	Rationale
PRDM1	up	Master regulator of plasma-cell differentiation (selected a priori; distal DARs only)
IRF4	up	Induces PRDM1; regulatory layer above it
SDC1	up	CD138, the defining plasmablast surface marker
PAX5	down	Guardian of B-cell identity; represses PRDM1
MS4A1	down	CD20, lost during differentiation
CIITA	down	MHC-II regulator, silenced in plasma cells

The reason for this particular set is the **PRDM1–PAX5 mutual repression circuit**. The two factors inhibit each other, and differentiation happens when that switch tips. Having both ends of the circuit in the target list means the analysis can describe a mechanism rather than just a list of changed genes.

6.3 What the target loci look like

Extracting every DAR near the six targets produced a table that turned out to be more interesting than expected.

Gene	Direction	DARs	Distance to TSS	Strongest log2FC
PRDM1	13 up, 1 down	14	94–235 kb	+3.54
IRF4	2 up, 6 down	8	1.8–107 kb	–3.10
SDC1	6 up	6	0–73 kb	+2.62
PAX5	6 down	6	1.2–198 kb	–2.80
MS4A1	1 down	1	**0**	**–3.17**
CIITA	1 up, 4 down	5	4–25 kb	–2.80

Three observations, each of which speaks to a different pitfall from the assignment.

MS4A1 is the clean case. A single peak sitting directly on the TSS, log2FC –3.17 with padj 3.9×10^{-10}. Roughly nine-fold less accessible in plasmablasts. This is what a textbook promoter shutdown looks like, and it needs no interpretation.

PRDM1 is the opposite, and the more informative one. Fourteen DARs in total, of which **13 of 14 gain accessibility in plasmablasts**; the single exception (peak_123043, $\log_2$FC –1.84) sits 117 kb from the TSS on the opposite side. None of the fourteen is near the promoter (the closest is 94 kb away and the furthest 235 kb). The plasma-cell master regulator is not switched on through its promoter in this dataset; it is surrounded by a field of distal elements, thirteen of which gain accessibility together. This is consistent with selective activation of candidate distal regulatory elements, though as §8.5 sets out, their assignment to PRDM1 rests on proximity rather than on measured interaction. It is also a direct demonstration of why pooling all peaks in a gene body would be misleading: averaging fourteen distal peaks (thirteen up, one down) against an unchanged promoter would dilute a striking signal into a weak one.

It also exposes the limit of nearest-gene assignment. Those thirteen peaks are nearest to PRDM1, but at 100–235 kb a peak can perfectly well loop to a different gene. Without chromatin interaction data I cannot resolve this, and I would not claim more than "consistent with PRDM1 regulation".

IRF4 and CIITA move in both directions at once. IRF4 has two up-DARs (one 1.8 kb from its TSS at +2.12, one 107 kb away at +1.08) against six down-DARs 32–99 kb away. CIITA is similar. A promoter opening while distal elements close is not a contradiction (it is what you would expect if a locus is being rewired rather than simply switched on or off). It does mean that any summary statistic per gene would erase the signal.

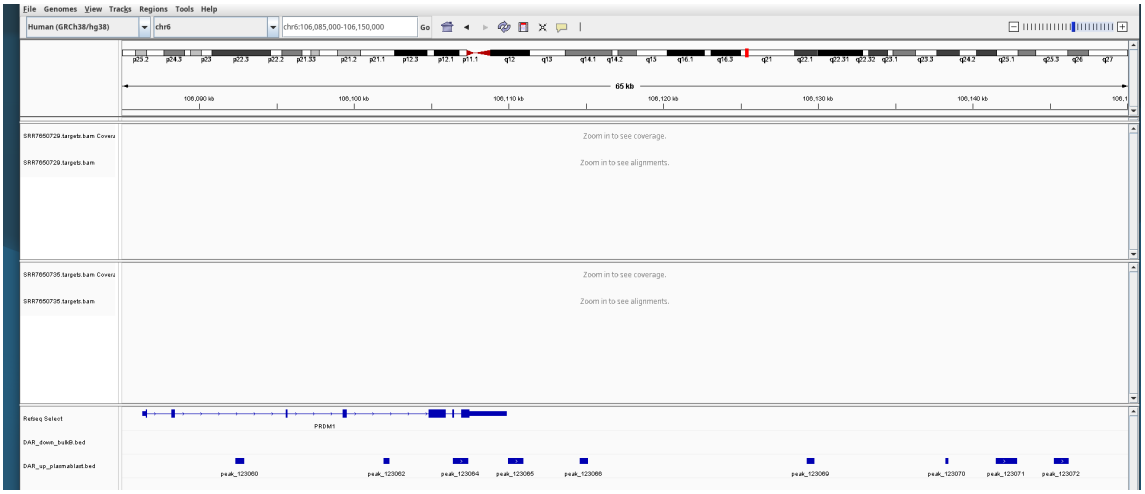
I also examined PAX5 in Week 2 by looking at raw counts across its gene body and got a confusing mixed picture. Section 6.1's distance-based approach explains why: the window I used then spanned the whole gene body, pooling elements that behave differently.

6.4 IGV inspection

Two loci were inspected visually in IGV, loading subsetted BAMs for one matched donor pair (SRR7650729, bulk B and SRR7650735, plasmablast, both from donor 1001) together with the two DAR track files.



(a) *MS4A1*, chr11:60,450,000–60,460,000. `peak_23690` from the down-DAR track sits directly over the promoter.



(b) *PRDM1*, chr6:106,085,000–106,150,000. Nine up-DARs span the locus; the down-DAR track is empty throughout.

Figure 5. IGV views of two target loci, one matched donor pair (SRR7650729 bulk B, upper tracks; SRR7650735 plasmablast, lower tracks). IGV suppresses read-coverage rendering above a 30 kb window, so panel (b) shows the peak tracks only.

Panel B4 makes the enhancer-field pattern visible in a way the table does not: a row of peaks spanning 65 kb, every one in the same direction, and nothing at all in the opposite direction.

A practical note: IGV suppresses coverage rendering above a 30 kb window by default, so the wide PRDM1 view shows the peak tracks but not the read pileups. The peak track is the informative layer at that scale.

6.5 Motif enrichment with HOMER

HOMER was run separately on the up and down DAR sets:

```
findMotifsGenome.pl DAR_up_plasmablast.bed $HOMER_REF homer_up/ \
  -bg consensus_all.bed -size 200 -p 4
```

Two parameter choices carry the analysis.

`-bg consensus_all.bed` replaces HOMER's default random-genomic background with the full consensus peak set. This changes the question being asked. Against random background, the answer would be "what do open chromatin regions contain" (mostly generic promoter features). Against the consensus set, the question becomes "what distinguishes regions that changed from regions that are open in both cell types", which is the biologically meaningful comparison.

`-size 200` restricts the search to ±100 bp around each peak centre. This is the assignment's "peak resolution" warning made concrete: transcription factor binding sites cluster near peak summits, and searching a full 759 bp median peak width would dilute the true motif into surrounding background.

The result is a clean, almost binary split between the two directions.

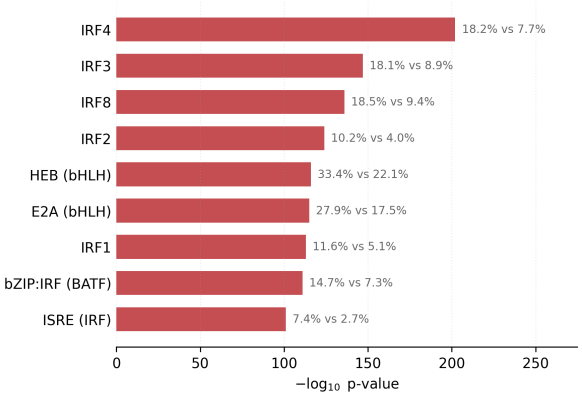
Opening in plasmablasts - the IRF family:

Motif	p-value	% targets	% background
IRF4	1e-202	18.22%	7.69%
IRF3	1e-147	18.14%	8.87%
IRF8	1e-136	18.48%	9.38%
IRF2	1e-124	10.16%	3.98%
HEB (bHLH)	1e-116	33.38%	22.15%
E2A (bHLH)	1e-115	27.92%	17.52%
IRF1	1e-113	11.55%	5.08%
bZIP:IRF (BATF)	1e-111	14.74%	7.35%
ISRE (IRF)	1e-101	7.41%	2.70%

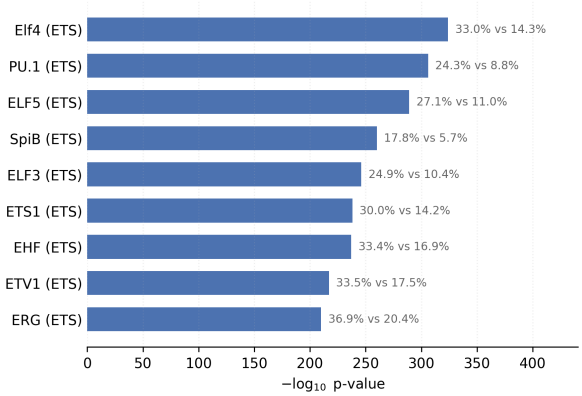
Staying open in bulk B cells - the ETS family:

Motif	p-value	% targets	% background
Elf4 (ETS)	1e-324	32.95%	14.27%
PU.1 (ETS)	1e-306	24.26%	8.82%
ELF5 (ETS)	1e-289	27.07%	10.97%
SpiB (ETS)	1e-260	17.75%	5.71%
ELF3 (ETS)	1e-246	24.89%	10.42%
ETS1 (ETS)	1e-238	29.99%	14.22%
EHF (ETS)	1e-237	33.43%	16.85%
ETV1 (ETS)	1e-217	33.50%	17.48%
ERG (ETS)	1e-210	36.88%	20.41%

Three things stand out.



(a) Regions opening in plasmablasts (dominated by the IRF family).



(b) Regions staying open in bulk B cells (dominated by the ETS family).

Figure 6. HOMER known-motif enrichment, background = full consensus peak set, `-size 200`. Labels give the percentage of target versus background sequences containing each motif.

IRF4 is the top motif in the up set. IRF4 is also one of my six target genes, chosen for entirely independent reasons in Section 6.2. So the gene-level analysis (IRF4's promoter gains accessibility) and the sequence-level analysis (IRF4 binding sites are the most enriched motif in newly opened regions) agree without being designed to.

PU.1 and SpiB dominate the down set. These are the ETS factors that hold B-cell identity in place. PU.1 appears in 24.3% of closing regions against 8.8% of background (a threefold enrichment at p = 1e-306). The picture is of an ETS-based regulatory network being dismantled and replaced by an IRF-based one.

The composite bZIP:IRF motif at 1e-111 is consistent with composite binding. IRF4 binds DNA weakly on its own and works through composite elements with partners such as BATF. Finding the composite motif enriched, and not only the bare IRF site, is consistent with functional IRF4–BATF composite binding, although motif enrichment alone cannot establish occupancy (that would require footprinting or ChIP data).

The bHLH motifs (HEB, E2A) appearing in the up set are worth a caveat. E2A also appears further down the down-set list, so it is probably a general B-lineage factor active in both states rather than something specific to the transition.

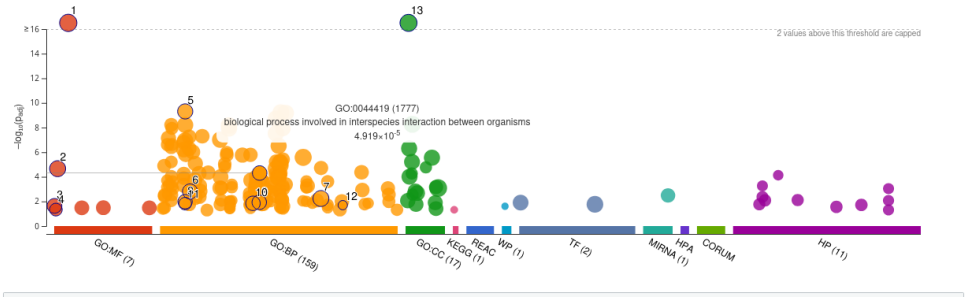
6.6 GO enrichment with g:Profiler

Three queries were run: the two promoter-proximal gene lists and the six target genes.

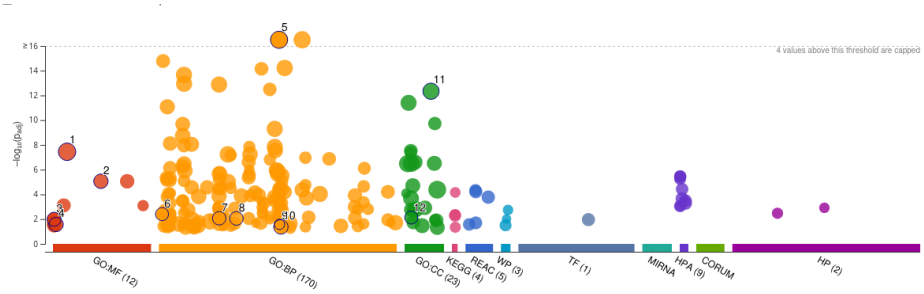
Bulk B (down) - the most specific result:

Term	Source	p-adj
cell activation	GO:BP	1.6e-15
lymphocyte activation	GO:BP	3.2e-13
B cell activation	GO:BP	1.4e-08
B cell proliferation	GO:BP	1.0e-06
Hematopoietic cell lineage	KEGG	6.9e-05
Factor: PU.1	TF	1.1e-02

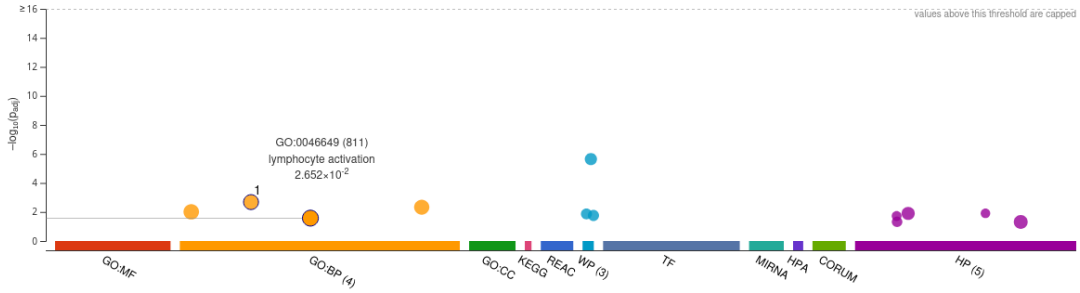
The last row deserves attention. g:Profiler's transcription-factor analysis independently identified **PU.1** as enriched in this gene set (the same factor HOMER found at 1e-306 in the same regions, using a completely



(a) Genes near regions opening in plasmablasts (686 mapped).



(b) Genes near regions staying open in bulk B cells (696 mapped).



(c) The six target genes.

Figure 7. g:Profiler functional enrichment. The x-axis groups terms by source database; orange = GO:BP. Numbered points are detailed in the accompanying result tables.

different method and database). HOMER scans DNA sequence under the peaks; g:Profiler looks up curated TF–target relationships for the gene names. Two orthogonal approaches converging on the same regulator is about as good as this kind of evidence gets.

Plasmablast (up): activation of immune response (2.4e-07), immune response-activating signalling pathway (1.2e-08), positive regulation of immune system process (5.9e-09). Broader and less specific than the down set, which is expected - the up set includes many effector and secretory genes that spread across immune categories rather than concentrating in one. A SMAD4 motif also appeared (TF, 1.7e-02), consistent with TGF-β involvement in plasma-cell differentiation.

The six target genes: despite only six inputs, the enrichment is sharp - lymphocyte differentiation (GO:BP, 2.1e-03) and, more specifically, Extrafollicular and follicular B cell activation (WikiPathways, 2.3e-06). Four of the six genes fall in each of these terms.

This should be read as a sanity check rather than as validation. The six targets were chosen precisely because they are canonical B-cell and plasma-cell markers (§6.2), so recovering B-cell differentiation terms from them is close to circular - the enrichment confirms that the annotation databases agree with the textbook biology that motivated the selection, not that the selection was independently correct. The genuinely independent checks are elsewhere: the promoter-proximal marker recovery in §6.2, the PRDM1

statistics in §5.6, and the convergence of HOMER and g:Profiler on PU.1 below.

7. Problems I Ran Into

Roughly two thirds of my time in Week 1 went into these problems rather than into running the intended commands, and I learned more from them than from the steps that worked.

7.1 Picard crashed on a missing read group

```
java.lang.NullPointerException: Cannot invoke
"htsjdk.samtools.SAMReadGroupRecord.getReadGroupId()"
because the return value of "SAMRecord.getReadGroup()" is null
```

A null-pointer exception from a Java tool is not a friendly error message, and my first thought was that my BAM was corrupted. It was not. Picard organises duplicate detection by library, identified through the `@RG` read-group tag on each alignment. Bowtie2 does not add read groups unless explicitly told to with `--rg-id` and `--rg`, so every record returned null.

The fix was `AddOrReplaceReadGroups` on the existing sorted BAM, avoiding a full re-alignment. Adding the read group during alignment would be cleaner practice, and is what nf-core does internally.

What I took from this: tool contracts are often implicit. Bowtie2 produces a perfectly valid BAM. Picard requires a more constrained BAM. Nothing in the protocol mentioned the dependency, and that gap is exactly what a curated workflow hides from you.

7.2 MACS2 would not run at all

```
ImportError: .../MACS2/IO/ScoreTrack.cpython-311-x86_64-linux-gnu.so:
undefined symbol: __log_finite
```

MACS2 contains compiled Cython modules. The build in the shared Conda environment was compiled against a glibc that exported `__log_finite`, an internal symbol glibc removed from its public interface at version 2.28. Davros runs Debian 13 with a much newer glibc, so the dynamic linker cannot resolve the symbol and the module never loads. The failure happens before any argument parsing, so no change to my command could have avoided it.

My attempt to install a working build into a private Conda environment was then blocked by an unaccepted Anaconda Terms-of-Service policy, adding another detour. In the end I used MACS3 3.0.2, already in the system path - the maintained successor to MACS2 with a nearly identical `callpeak` interface, needing no change beyond the executable name.

This is a textbook reproducibility failure and the thing I would most want to warn the next person about. I pinned the software version exactly as specified. My parameters were correct. My input was valid. And it still could not run, because a binary artefact was incompatible with a system library (a dependency completely invisible at the level of a package specification). Containerisation, which is what the Docker-based nf-core pipeline provides and what this exercise deliberately bypassed, exists to eliminate exactly this class of problem. Hitting it myself made that argument far more concrete than reading about it would have.

7.3 The wrong `picard`

Before finding the real problem above, I ran `picard` without activating the Conda environment. It launched an unrelated system program of the same name (a graphical music-file tagger) which tried to open an X11 display and aborted with a Qt error. I spent longer than I would like to admit wondering why a bioinformatics tool wanted a graphics display. The lesson is trivial but useful: check `which <tool>` before blaming the tool.

7.4 Missing logs and the wrong MultiQC

My first Cutadapt run wrote only the trimmed FASTQ files - I never redirected its stdout summary to a file, so MultiQC had nothing to parse and the trimming statistics were simply absent from the report. I had to re-run the command with output captured.

Separately, one MultiQC run used the system version 1.21 instead of the environment's 1.35, producing a report with missing JavaScript assets and non-functional interactive plots. Same root cause as §6.3.

Both are small mistakes but together they cost an afternoon, and both come down to one habit: check what you are actually running, and capture logs at the moment you run something rather than trying to reconstruct them later.

7.5 HOMER had no genome installed, and I could not install one

`annotatePeaks.pl` silently produced an empty file. The reason was in the log I had not checked: HOMER ships without genome data, and the shared environment had none installed. Running `configureHomer.pl -install hg38` then produced thousands of lines of `print() on closed filehandle CONFIGOUT` (the installer could not write to the shared directory because I do not own it).

The supervisor resolved this by placing a reference FASTA on the server, which `findMotifsGenome.pl` accepts in place of a genome name. That works because HOMER only needs the sequence under the peaks, not the full annotation package.

Two things I would do differently. First, check the log immediately rather than assuming an empty output means an empty result (the same mistake as §7.4). Second, `configureHomer.pl` failing thousands of times without a single clear "permission denied" is poor error reporting, but the first line of the output did say what was wrong, and I scrolled past it.

7.6 A column name, and why the assertion mattered

When I rebuilt the metadata table from SraRunTable.csv, the donor column came through as `Donor` with a capital D, while my earlier QC script referred to `donor`. The PCA step failed with `the argument 'intgroup' should specify columns of colData(dds)`.

Trivial to fix, but worth recording for the opposite reason to the others: it failed loudly. The same class of mistake in the sample ordering (count columns not matching metadata rows) would have failed silently and produced a beautifully formatted analysis of the wrong comparison. That is why the `stopifnot` in §5.1 is in the script.

8. Discussion

8.1 The duplication rate is real

At 76.59%, duplication is the defining feature of my Week 1 sample. Before the reference report arrived I had two competing explanations and no way to choose:

1. **Genuine low library complexity** (too little starting material or too many PCR cycles, so few distinct molecules were amplified many times).
2. **An artefact of subsetting** - since duplicate detection is coordinate-based, restricting reads to one chromosome might increase coincidental collisions and inflate the rate.

The reference settled it. nf-core processed the same library across the whole genome at full depth and measured **73.76%** (within three points of my 76.59%). If subsetting were inflating the number, the whole-genome value would be much lower. It is not. The high duplication is a real property of this library.

I find this satisfying because a perfectly reasonable hypothesis was falsified by independent data. That is what having a reference is for.

The practical consequence is lost sensitivity, not lost specificity. The peaks I called are trustworthy; the concern is the ones I missed (exactly the pattern in the 90% recovery figure of §3.3).

8.2 Why a low-complexity library can still have excellent signal

Section 3.4 showed my sample with the worst duplication and the best FRiP in the project. These point in opposite directions only if you assume they measure the same thing:

- **Duplication** measures library complexity (how many distinct molecules were captured). A question of quantity.
- **FRiP** measures signal-to-noise (what fraction of reads fall inside peaks). A question of quality.

A library can be low-complexity and highly specific at once. The most plausible reading is that tagmentation in this sample was very selective, concentrating insertions in genuinely accessible chromatin, but input material was limited, so few distinct fragments were produced and PCR had to work hard. The result is a small set of unusually informative molecules, sequenced redundantly. A FRiP above 0.3 is generally considered high quality; 0.554 is well above that.

8.3 The effective genome size is an approximation

`-g 4.0e7` was estimated, not measured. Chromosome 22 is 50,818,468 bp in hg38, of which roughly 10–11 Mb is acrocentric short arm (satellite repeat and rDNA that short reads cannot map). Around 40 Mb is defensible but not rigorous. A better approach would use a mappability track at the relevant read length. Re-running at `-g 3.5e7` and `-g 4.5e7` and seeing how the peak count responds would show how much the choice actually matters, and I would like to do that before final submission.

8.4 What 14,662 DARs means biologically

Finding 14.4% of tested consensus peaks differentially accessible is a lot, but the comparison justifies it. Bulk B cells and plasmablasts are not two conditions of the same cell - they are two stages of a differentiation programme that involves shutting down one transcriptional identity and installing another. The near-balance between the two directions (7,976 opening, 6,686 closing) fits that picture: differentiation is not simply "opening more chromatin", it is a coordinated exchange.

The PRDM1 result in §5.6 shows what this looks like at a single locus (selective activation of particular elements rather than a uniform shift). Whether that generalises is a Week 3 question, and HOMER motif enrichment on the two DAR sets is the natural way to ask it.

8.5 The limits of nearest-gene assignment

Everything in Section 6 that connects a peak to a gene rests on one assumption: that a regulatory element controls the gene whose TSS is closest. For promoter-proximal peaks this is safe. For PRDM1's thirteen plasmablast-opening DARs at 94–235 kb it is a guess.

Enhancers routinely skip over intervening genes to reach their targets through chromatin looping, and distance is a poor predictor of which gene an element regulates. The honest position is that the PRDM1 enhancer field is consistent with PRDM1 regulation - the direction, the coordination across thirteen elements, and the known biology all point that way (but not demonstrated by this data alone). Resolving it would need Hi-C, promoter capture, or eQTL data.

This is also why I ran GO enrichment on the 5 kb promoter-proximal lists rather than the unrestricted nearest-gene lists, even though the latter are five times larger. Adding 3,500 uncertain gene assignments would have inflated the input without improving the evidence.

8.6 Why bidirectional loci matter

IRF4 and CIITA both have DARs in both directions (a promoter or promoter-proximal element opening while distal elements close). This looked like a contradiction when I first saw it.

It is not. A locus being rewired rather than simply switched will show exactly this: one set of regulatory inputs disengaging while another engages. What it does mean is that any analysis summarising accessibility per gene (averaging peaks, taking a maximum, counting DARs without regard to direction) would erase the signal or, worse, report a misleading average near zero.

The practical lesson for the assignment's "intrinsic enhancers" question is that peaks should be analysed as individual elements, and pooled only when there is a reason to believe they act together.

8.7 What the fragment-size distribution said about nucleosomes

The insert-size distribution in Figure 1 D1 is the direct answer to the assignment's nucleosome-positioning question. The library shows the expected periodicity: a dominant sub-nucleosomal population at ~45 bp (44.3% of fragments), a mono-nucleosomal mode at ~210 bp, and a di-nucleosomal mode at ~354 bp.

The spacing between modes (roughly 165 bp) corresponds to the DNA occupied by one nucleosome plus linker. This is a physical measurement of chromatin packaging read directly out of fragment lengths. To be precise about the alignment setting: the periodic pattern itself lies largely below 500 bp and would have been visible at Bowtie2's default insert limit; what `-X 2000` additionally preserves is the higher-order nucleosomal tail beyond 500 bp, which is 8.8% of fragments (§2.8). The distribution also has a practical consequence for motif analysis: transcription factor binding happens in the nucleosome-free fraction, which is why restricting HOMER to ±100 bp around peak summits (§6.5) is the right choice.

8.8 Deviations from the source protocol and the slides

Aspect	Reference / instructions	Mine	Why
Aligner (vs. nf-core)	BWA-MEM 0.7.17	Bowtie2	Specified by the module instructions
Genome build	hg19 (paper)	GRCh38	Required for the supplied hg38 blacklist
Cutadapt settings	min length 20, overlap 5	defaults	Measured impact: 7 read pairs
chrM removal	performed	not applicable	Input was chr22 only
Peak caller	MACS2 2.2.7.1	MACS3 3.0.2	MACS2 non-functional (§7.2)
Effective genome	whole genome	4.0×10^7	Input was chr22 only
Filter threshold	N = half of samples (PDF)	N = smallest group (slides)	Better rationale; both tested (§5.2)
DA design	`~ Donor + cell_type`	`~ cell_type` + paired sensitivity	Only 2 of 5 donors matched (§5.3)
HOMER genome	installed genome package	reference FASTA passed directly	No write access to the shared HOMER directory (§7.5)
HOMER background	random genomic (default)	full consensus peak set	Asks the biologically meaningful question (§6.5)
Alignment filter	MAPQ 30, `-F 1804 -f 2`	identical	-
Max insert	2 kb	identical	-

8.9 The g:Profiler background

One methodological choice in §6.6 deserves flagging. All three g:Profiler queries were run against the tool's default background, which is the full annotated genome. A more rigorous universe would have been the set of genes that could have been detected in this experiment (genes carrying a promoter-proximal peak in the consensus set, whether or not that peak changed). Restricting the background that way removes an implicit bias: a gene lying in closed chromatin can never enter the query list, so testing against all annotated genes slightly inflates the apparent enrichment of anything active in this lineage.

g:Profiler supports custom backgrounds directly, so this would have been straightforward. The effect is most likely to inflate broad immune categories rather than to manufacture specific ones, and the strongest results here (B cell receptor signalling, B cell activation, the PU.1 TF term) are specific rather than broad. I do not think the conclusions change, but it is a shortcut rather than best practice.

8.10 Limitations

1. **One sample, one chromosome** in Week 1 (no replicate, so no variance estimate, and chromosome 22 is gene-dense and GC-rich rather than representative).
2. **Low effective depth** in Week 1 (346,044 unique fragments is below where the ATAC-seq detection curve saturates).
3. **No TSS enrichment metric** (arguably the most informative ATAC-seq quality measure, and I did not calculate it).
4. **Incomplete donor pairing** limits the design (§5.3). The sensitivity analysis in §5.7 shows the conclusions are robust to donor adjustment as far as two matched donors can test, but this does not formally exclude residual confounding.
5. **Nearest-TSS gene assignment is an assumption, not a measurement** (§8.5). This is the weakest link in the Week 3 conclusions, particularly for PRDM1's distal elements.
6. **No chromatin interaction data**, so enhancer-to-gene links cannot be verified.
7. **The BED-to-SAF conversion carried a one-base offset** (§4.2). It is systematic across all peaks and samples, so it cannot bias the between-group comparison, but it is an error in the code as written.
8. **GO enrichment used the default genome background** rather than a peak-derived universe (§8.9).
9. **HOMER *de novo* motif discovery was not completed** (only the known-motif scan is reported). The known-motif result answers the question asked (which characterised TFs are enriched), but novel motifs were not explored.

9. Conclusions

The pipeline works, and I can show that rather than assert it. The manual Week 1 pipeline and the nf-core reference, run on the same library with a different aligner and a different peak-caller version, agreed on duplication to within three percentage points (76.59% vs 73.76%) and, at the end of the pipeline, on the usable fraction of the input to within half a point (20.21% vs 20.73%). A rough cross-lineage scaling placed the chromosome-22 peak count at about 90% of the estimated expectation (§3.3 sets out why this is an approximation rather than a precise recovery figure), and the FRiP score was 0.574 against the reference's 0.554 at broadly comparable per-chromosome depth.

A shared server is not automatically faster. Pinned to two threads on identical input, davros completed the alignment in 5m02s against 5m49s on a 2012-era Core i3 desktop (15.5% faster, considerably less than I expected). The user/real ratio was 2.03 on every run, so both machines parallelised equally well and the gap reflects per-core throughput alone. A server's advantage lies in aggregate throughput and memory, and constraining a job to two threads removes almost all of it.

Bulk B cells and plasmablasts differ enormously in chromatin accessibility. PC1 accounts for 87% of the variance and separates the two cell types cleanly. 14,662 of the 101,592 testable consensus peaks are

differentially accessible, roughly balanced between opening (7,976) and closing (6,686). Differentiation is a coordinated exchange, not a general opening of chromatin.

The main conclusions appear robust to donor adjustment, and I tested this rather than assuming it. The paired design on the four matched samples recovered 3,696 DARs, of which 3,692 (99.9%) were also found by the main analysis, with complete agreement on the direction of change. This is a strong concordance, though it does not formally exclude residual donor confounding (the test is limited to the two donors that contributed both cell types).

The regulatory logic is an ETS-to-IRF handover. Regions closing in plasmablasts are dominated by ETS motifs, above all PU.1 (1e-306) and SpiB (1e-260) (the factors that hold B-cell identity in place). Regions opening are dominated by the IRF family, with IRF4 top at 1e-202. The composite bZIP:IRF motif (1e-111) suggests these are functional IRF4–BATF complexes rather than isolated sites.

Three independent lines of evidence converge. IRF4 was selected as a target gene from promoter-proximal DARs; it is also the top enriched motif in the up set. PU.1 was found by HOMER through DNA sequence scanning and independently by g:Profiler through curated TF–target annotation. Neither convergence was designed in.

Individual loci show that regulation is element-specific, not gene-specific. MS4A1 loses accessibility at a single promoter peak (log2FC –3.17). PRDM1 gains it across thirteen distal elements 94–235 kb from the TSS, with nothing at the promoter. IRF4 and CIITA move in both directions at once. Any per-gene summary statistic would have erased all three patterns.

Week 1 and Weeks 2–3 used different data, and separating them matters. SRR7650763 is an NK library, so Week 1 is a methods exercise and Section 3 is a technical validation of the pipeline, not evidence about B-cell biology. All biological claims in this report rest on the seven B-lineage samples of Weeks 2 and 3.

The technical failures were as instructive as the results. A missing read group and a glibc symbol incompatibility both cost hours, and both are exactly the class of problem containerised workflows exist to prevent. Having pinned the correct software version and still been unable to run it is a more convincing argument for nf-core than any amount of reading about reproducibility.

Limitations, restated

The nearest-TSS assignment is the weakest link in Week 3. PRDM1's thirteen distal peaks are attributed to PRDM1 because it is nearest, but at 100–235 kb they could loop elsewhere. Without chromatin interaction data this cannot be resolved, and the enhancer-to-gene assignments should be read as plausible rather than established.

Only two of five donors contributed both cell types, which is a property of the source dataset rather than of the sample selection.

If this work were continued

The analysis is complete as submitted. Three things would strengthen it further, in rough order of value:

1. **An effective-genome-size sensitivity check** (§8.3). Re-running MACS3 at `-g 3.5e7` and `-g 4.5e7` would show how much the estimated 40 Mb figure actually influences the Week 1 peak count. If the number is stable across that range, the choice can be dismissed as inconsequential rather than defended.
2. **Chromatin interaction data** to test the enhancer-to-gene assignments (§8.5), particularly the thirteen distal PRDM1 peaks. Promoter-capture Hi-C would turn a plausible attribution into a demonstrated one.

3. **TSS enrichment** as an additional Week 1 QC metric, which captures signal-to-noise more directly than alignment rate or duplication.

Appendix A - Scripts

All scripts are reproduced as run. Shell work was done in `~/raw_fastq` (Week 1) and `~/week2` (Week 2).

A.1 Week 1 - shell pipeline

```
#!/bin/bash
# Week 1: manual ATAC-seq pre-processing of SRR7650763 (chr22 subset)

# --- Environment ---
source /pcpool/home/sysbiol_tutor/miniconda3/bin/activate
conda activate /pcpool/home/sysbiol_share/envs/atac
cd ~/raw_fastq

# --- Step 1: adapter trimming (Nextera/Tn5 mosaic end) ---
cutadapt \
  -a CTGTCTCTTATACACATCT \
  -A CTGTCTCTTATACACATCT \
  -o output_cutadapt/trimmed_R1.fastq.gz \
  -p output_cutadapt/trimmed_R2.fastq.gz \
  SRR7650763_1_chr22.raw.fastq.gz \
  SRR7650763_2_chr22.raw.fastq.gz \
  > output_cutadapt/cutadapt.log

# --- Step 2: alignment. -X 2000 preserves fragments >500 bp (8.8% here) ---
# Prefix with `time` to capture wall-clock runtime for the benchmark
bowtie2 -p 2 -X 2000 \
  -x ref/GRCh38_chr22 \
  -1 output_cutadapt/trimmed_R1.fastq.gz \
  -2 output_cutadapt/trimmed_R2.fastq.gz \
  -S sample.sam \
  2> SRR7650763_bowtie2.log

# --- Step 3a: quality and flag filtering (ENCODE standard) ---
# -q 30  : MAPQ >= 30, removes low-confidence and multi-mapping reads
# -f 2   : keep properly paired reads only
# -F 1804: drop unmapped(4) + mate unmapped(8) + secondary(256)
#          + vendor QC fail(512) + flagged duplicate(1024)
samtools view -b -q 30 -F 1804 -f 2 sample.sam > sample.filtered.bam

# --- Step 3b: coordinate sort (required by Picard) ---
samtools sort -@ 2 -o sample.sorted.bam sample.filtered.bam

# --- Step 3c: add read group (Picard requires @RG; Bowtie2 does not add one) ---
picard AddOrReplaceReadGroups \
  I=sample.sorted.bam \
  O=sample.rg.bam \
  RGID=1 RGLB=lib1 RGPL=ILLUMINA RGPU=unit1 RGSM=SRR7650763

# --- Step 3d: remove PCR duplicates ---
picard MarkDuplicates \
  I=sample.rg.bam \
  O=sample.dedup.bam \
  M=SRR7650763.dup_metrics.txt \
  REMOVE_DUPLICATES=true

samtools index sample.dedup.bam

# --- Blacklist filtering. Verify genome build first! ---
```

```
samtools view -H sample.dedup.bam | grep '^@SQ'    # expect SN:22 LN:50818468 (hg38)
head -3 hg38-blacklist.ensembl.bed                 # expect Ensembl naming (no 'chr')

bedtools intersect -v \
  -abam sample.dedup.bam \
  -b hg38-blacklist.ensembl.bed \
  > sample.clean.bam
samtools index sample.clean.bam

# --- Step 4: peak calling ---
# -f BAMPE   : use real paired-end fragment coordinates
# -g 4.0e7   : effective genome size for chr22 only (NOT whole-genome 'hs')
# --keep-dup all : duplicates already removed by Picard; do not filter twice
macs3 callpeak \
  -t sample.clean.bam \
  -f BAMPE \
  -g 4.0e7 \
  -n SRR7650763 \
  --broad --broad-cutoff 0.1 \
  --keep-dup all \
  --outdir macs2_out \
  2> macs2.log

# --- Insert-size distribution (key ATAC-seq QC metric) ---
picard CollectInsertSizeMetrics \
  I=sample.dedup.bam \
  O=SRR7650763.insert_size_metrics.txt \
  H=SRR7650763.insert_size_histogram.pdf \
  M=0.5

# --- Benchmark: same command, two threads, both platforms ---
# Output to /dev/null so disk write speed does not contaminate the timing
time bowtie2 -p 2 -X 2000 -x ref/GRCh38_chr22 \
  -1 output_cutadapt/trimmed_R1.fastq.gz \
  -2 output_cutadapt/trimmed_R2.fastq.gz \
  -S /dev/null 2> /dev/null

# --- FRiP score (fraction of reads in peaks) ---
IN=$(bedtools intersect -a sample.clean.bam \
       -b macs2_out/SRR7650763_peaks.broadPeak -u | samtools view -c)
TOT=$(samtools view -c sample.clean.bam)
echo "scale=4; $IN/$TOT" | bc      # 396476 / 690686 = 0.5740

# --- Step 5: aggregate all logs into one report ---
multiqc . -o multiqc_out --force

# --- Verification ---
samtools view -c sample.dedup.bam                  # 692,088
samtools view -c sample.clean.bam                  # 690,686
wc -l macs2_out/SRR7650763_peaks.broadPeak         # 1,035
awk '{print $3-$2}' macs2_out/SRR7650763_peaks.broadPeak | sort -n | \
  awk '{a[NR]=$1} END {print "min:",a[1], "median:",a[int(NR/2)], "max:",a[NR]}'
grep -A1 "LIBRARY" SRR7650763.dup_metrics.txt | cut -f1,3,9
```

A.2 Week 2 Days 1–2 - consensus peaks and feature counts

```
#!/bin/bash
# Week 2 Days 1-2: consensus peak set and fragment counting (7 samples)

source /pcpool/home/sysbiol_tutor/miniconda3/bin/activate
conda activate /pcpool/home/sysbiol_share/envs/atac

mkdir -p ~/week2 && cd ~/week2
```

```
export PEAK_DIR=/pcpool/home/sysbiol_tutor/Projects_Haubrock/student_projects/sysbiol19/P04_Bulk_B_vs_Plasmablas
export BAM_DIR=$PEAK_DIR
export OUT_DIR=~/week2

# Build the sample list from the files themselves (avoids typos)
SAMPLES=($(ls $PEAK_DIR/*_peaks.broadPeak | xargs -n1 basename | sed 's/_REP1.*//'))
echo "Samples (${#SAMPLES[@]}): ${SAMPLES[@]}"

# --- Consensus peak set ---
# Extract coordinates from every sample, pool, sort, merge overlaps.
# Peaks unique to one sample are retained; other samples get near-zero counts.
for sample in "${SAMPLES[@]}"; do
    awk '{print $1"\t"$2"\t"$3}' "${PEAK_DIR}/${sample}_REP1.mLb.clN_peaks.broadPeak"
done | sort -k1,1 -k2,2n | bedtools merge -i - > "${OUT_DIR}/consensus_peaks.bed"

wc -l consensus_peaks.bed    # 155,948

# --- Convert BED to SAF, assigning a unique ID per region ---
# NOTE: BED is 0-based half-open, SAF is 1-based inclusive -> start should be $2+1.
# Reproduced here as actually run; see section 4.2 for the impact assessment.
awk 'BEGIN{OFS="\t"; print "GeneID","Chr","Start","End","Strand"}
     {print "peak_"NR,$1,$2,$3,"."}' consensus_peaks.bed > consensus_peaks.saf

# --- Build the BAM list in the SAME order as SAMPLES ---
BAMS=()
for sample in "${SAMPLES[@]}"; do
    BAMS+=("${BAM_DIR}/${sample}_REP1.mLb.clN.sorted.bam")
done

# Verify all BAMs and their indices exist before counting
for b in "${BAMS[@]}"; do
    [[ -f "$b" ]]        || { echo "ERROR: BAM not found: $b"; exit 1; }
    [[ -f "${b}.bai" ]]  || { echo "ERROR: index not found: $b"; exit 1; }
done

# --- Fragment counting ---
# -p --countReadPairs : count fragments, not individual reads
# -B : count only fragments with both ends successfully aligned
# -C : discard chimeric pairs (ends on different chromosomes)
featureCounts \
  -F SAF -a consensus_peaks.saf \
  -o "${OUT_DIR}/counts_matrix.txt" \
  -p --countReadPairs -B -C -T 4 \
  "${BAMS[@]}"

cat counts_matrix.txt.summary

# --- Clean the matrix: keep peak ID + count columns, rename headers ---
# featureCounts columns: 1=GeneID 2=Chr 3=Start 4=End 5=Strand 6=Length 7-13=samples
awk 'NR==1 && /^#/ {next}
     NR==2 {print "PeakID\tSRR7650729\tSRR7650735\tSRR7650771\tSRR7650777\tSRR7650812\tSRR7650851\tSRR7650918";
     {print $1"\t"$7"\t"$8"\t"$9"\t"$10"\t"$11"\t"$12"\t"$13}' \
     counts_matrix.txt > counts_clean.txt
```

A.3 `00_make_clinical_table.R` - metadata from SraRunTable

```
# Build the clinical metadata table from the official SRA run table.
# The CSV contains quoted fields with embedded commas, so read.csv is used
# rather than awk -F, which would split those fields incorrectly.

sra <- read.csv("SraRunTable.csv", stringsAsFactors = FALSE)
cat("SraRunTable rows:", nrow(sra), "\n")
```

```
# Take the sample order from the count matrix header. This order is
# authoritative: DESeq2 requires colData rows to match count columns exactly.
counts_header <- strsplit(readLines("counts_clean.txt", n = 1), "\t")[[1]]
samples <- counts_header[-1]
cat("Count matrix samples:", paste(samples, collapse = ", "), "\n\n")

# Subset to our samples, in that order, keeping only relevant columns
clin <- sra[match(samples, sra$Run),
            c("Run", "Donor", "cell_type", "lineage", "treatment")]

# Prefix donor IDs with a letter (R factor levels should not start with a digit)
clin$Donor     <- paste0("D", clin$Donor)
clin$cell_type <- ifelse(clin$cell_type == "Plasmablasts", "Plasmablast", clin$cell_type)
colnames(clin)[1] <- "sample"

print(clin)

# Integrity checks
stopifnot(all(clin$sample == samples))
stopifnot(!any(is.na(clin$cell_type)))
cat("\n>>> Order and integrity checks PASSED\n")
cat("Treatment (should be single):", unique(clin$treatment), "\n")
cat("Lineage (should be single):", unique(clin$lineage), "\n")
cat("Group sizes:\n");        print(table(clin$cell_type))
cat("Donor x cell type:\n");  print(table(clin$Donor, clin$cell_type))

write.table(clin, "clinical_table.txt", sep = "\t", quote = FALSE, row.names = FALSE)
cat("\n>>> clinical_table.txt written\n")
```

A.4 `01_load_and_filter.R` - SummarizedExperiment and low-count filter

```
suppressMessages({ library(SummarizedExperiment); library(DESeq2) })

counts  <- as.matrix(read.delim("counts_clean.txt", row.names = 1, check.names = FALSE))
coldata <- read.delim("clinical_table.txt", row.names = 1)

# Critical: count columns and metadata rows must be in identical order.
# A mismatch here would silently compare the wrong groups.
stopifnot(all(colnames(counts) == rownames(coldata)))
cat(">>> Column order match VERIFIED\n")

# Bulk_B as reference level => positive log2FC means "more open in plasmablasts"
coldata$cell_type <- factor(coldata$cell_type, levels = c("Bulk_B", "Plasmablast"))
coldata$Donor     <- factor(coldata$Donor)

se <- SummarizedExperiment(assays = list(counts = counts), colData = coldata)

# Low-count filter: >=10 counts in at least N samples.
# N = size of the smallest experimental group, so a peak present throughout
# one entire cell type is retained even if absent from the other.
N <- min(table(coldata$cell_type))
cat("Smallest group size -> N =", N, "\n")

keep <- rowSums(assay(se, "counts") >= 10) >= N
cat("Before filter:", nrow(se), " after:", sum(keep),
    sprintf(" (%.1f%% removed)\n", 100 * (nrow(se) - sum(keep)) / nrow(se)))

se <- se[keep, ]
saveRDS(se, "se_filtered.rds")
cat(">>> se_filtered.rds saved\n")
```

A.5 `02_deseq_qc.R` - model fitting, VST and exploratory QC

```
suppressMessages({
  library(DESeq2); library(SummarizedExperiment)
  library(vsn); library(hexbin); library(pheatmap)
  library(RColorBrewer); library(ggplot2)
})

se <- readRDS("se_filtered.rds")
dir.create("qc", showWarnings = FALSE)

# --- Design A: ~ cell_type on all 7 samples (main analysis) ---
# A paired design (~ Donor + cell_type) is not used here because only 2 of the
# 5 donors contributed both cell types; see the sensitivity analysis instead.
ddsA <- DESeqDataSet(se, design = ~ cell_type)
ddsA <- DESeq(ddsA)
cat(">>> Design A fitted (7 samples, ~ cell_type)\n")
cat("Size factors (library-depth normalisation):\n")
print(round(sizeFactors(ddsA), 3))

# --- Variance-stabilising transformation ---
# blind = FALSE lets the transformation account for the experimental design
vsdA <- vst(ddsA, blind = FALSE)

# --- meanSdPlot before and after VST ---
# The red running-mean line should flatten after transformation
png("qc/meanSd_before_vst.png", 900, 700, res = 120)
print(meanSdPlot(log2(counts(ddsA, normalized = TRUE) + 1), plot = FALSE)$gg +
      ggtitle("meanSdPlot: before VST (log2 normalised counts)"))
dev.off()

png("qc/meanSd_after_vst.png", 900, 700, res = 120)
print(meanSdPlot(assay(vsdA), plot = FALSE)$gg +
      ggtitle("meanSdPlot: after VST"))
dev.off()
cat(">>> meanSdPlot saved\n")

# --- PCA: do samples cluster by cell type or by donor? ---
pcaData <- plotPCA(vsdA, intgroup = c("cell_type", "Donor"), returnData = TRUE)
pv <- round(100 * attr(pcaData, "percentVar"))
png("qc/pca.png", 1000, 800, res = 120)
print(ggplot(pcaData, aes(PC1, PC2, color = cell_type, shape = Donor)) +
        geom_point(size = 5) +
        xlab(paste0("PC1: ", pv[1], "% variance")) +
        ylab(paste0("PC2: ", pv[2], "% variance")) +
        ggtitle("PCA (VST) - 7 samples") + theme_bw(base_size = 14))
dev.off()
cat(">>> PCA: PC1 =", pv[1], "%  PC2 =", pv[2], "%\n")

# --- Sample-to-sample distance heatmap ---
sampleDists <- dist(t(assay(vsdA)))
mat  <- as.matrix(sampleDists)
labs <- paste(vsdA$cell_type, vsdA$Donor, sep = " | ")
rownames(mat) <- labs; colnames(mat) <- labs
png("qc/sample_distance_heatmap.png", 900, 800, res = 120)
pheatmap(mat,
         clustering_distance_rows = sampleDists,
         clustering_distance_cols = sampleDists,
         col = colorRampPalette(rev(brewer.pal(9, "Blues")))(255),
         main = "Sample-to-sample distance (VST)")
dev.off()
cat(">>> Distance heatmap saved\n")

saveRDS(ddsA, "ddsA.rds"); saveRDS(vsdA, "vsdA.rds")
```

```
cat(">>> ddsA.rds, vsdA.rds saved\n")
```

A.6 `03_da_test.R` - differential accessibility testing

```
suppressMessages({ library(DESeq2); library(apeglm); library(EnhancedVolcano) })

ddsA <- readRDS("ddsA.rds")

# --- Contrast: Plasmablast vs Bulk_B (reference) ---
# Positive log2FC = region MORE open in plasmablasts
res <- results(ddsA, contrast = c("cell_type", "Plasmablast", "Bulk_B"))
cat("=== summary(res) ===\n"); summary(res)

cat("\nresultsNames:\n"); print(resultsNames(ddsA))

# --- log2FC shrinkage with apeglm ---
# Low-count peaks show inflated fold changes from sampling noise alone.
# apeglm shrinks those toward zero while leaving well-supported peaks intact.
resS <- lfcShrink(ddsA, coef = "cell_type_Plasmablast_vs_Bulk_B", type = "apeglm")
cat(">>> apeglm shrinkage applied\n")

# --- DAR criteria: FDR < 0.05 AND |log2FC| >= 1 ---
df <- as.data.frame(resS); df$PeakID <- rownames(df)
df <- df[!is.na(df$padj), ]
dars <- df[df$padj < 0.05 & abs(df$log2FoldChange) >= 1, ]
up   <- dars[dars$log2FoldChange >=  1, ]   # open in plasmablasts
down <- dars[dars$log2FoldChange <= -1, ]   # open in bulk B

cat("\n=== DAR SUMMARY (FDR<0.05 & |log2FC|>=1) ===\n")
cat("Peaks tested:", nrow(df), "\n")
cat("Total DARs:",   nrow(dars), "\n")
cat("  More open in plasmablasts:", nrow(up), "\n")
cat("  More open in bulk B      :", nrow(down), "\n")

# --- Volcano plot ---
png("qc/volcano.png", 1100, 950, res = 120)
print(EnhancedVolcano(df, lab = df$PeakID, x = "log2FoldChange", y = "padj",
      title    = "DA Analysis: Plasmablast vs Bulk B",
      subtitle = "apeglm-shrunk LFC, FDR < 0.05",
      pCutoff = 0.05, FCcutoff = 1,
      xlab = bquote(~Log[2]~"fold change"),
      ylab = bquote(~-Log[10]~italic(FDR)),
      pointSize = 1.5, labSize = 3.0, colAlpha = 0.6,
      legendPosition = "top"))
dev.off()
cat(">>> volcano.png saved\n")

# --- MA plot ---
png("qc/ma_plot.png", 950, 750, res = 120)
plotMA(resS, ylim = c(-6, 6), main = "MA plot (apeglm-shrunk)")
dev.off()

# --- Save results ---
df <- df[order(df$padj), ]
write.table(df, "DA_results_all.tsv", sep = "\t", quote = FALSE, row.names = FALSE)
write.table(dars[order(dars$padj), ], "DARs_significant.tsv",
            sep = "\t", quote = FALSE, row.names = FALSE)
saveRDS(resS, "resShrink.rds")

cat("\nTop 10 DARs:\n")
print(head(df[, c("PeakID", "baseMean", "log2FoldChange", "padj")], 10))
```

A.7 `04_validation_sensitivity.R` - biological validation and paired-design check

```
suppressMessages({ library(DESeq2); library(apeglm) })

se   <- readRDS("se_filtered.rds")
resS <- readRDS("resShrink.rds")
df   <- as.data.frame(resS); df$PeakID <- rownames(df)

# --- 1. Biological validation: PRDM1 locus (chr6) ---
# PRDM1/BLIMP1 is the master regulator of plasma-cell differentiation,
# so this locus should gain accessibility in plasmablasts.
prdm1 <- paste0("peak_", 123058:123067)
cat("=== PRDM1 locus peaks ===\n")
v <- df[rownames(df) %in% prdm1, c("baseMean", "log2FoldChange", "padj")]
v$DAR <- ifelse(!is.na(v$padj) & v$padj < 0.05 & abs(v$log2FoldChange) >= 1, "DAR", "-")
print(round(v[, 1:3], 3)); print(table(v$DAR))

# --- 2. Sensitivity analysis: true paired design on matched donors only ---
# Only D1001 and D1002 contributed both cell types.
matched <- colData(se)$Donor %in% c("D1001", "D1002")
seB <- se[, matched]
colData(seB)$Donor <- droplevels(colData(seB)$Donor)
cat("\n=== Design B: ~ Donor + cell_type ===\n")
print(table(colData(seB)$Donor, colData(seB)$cell_type))

ddsB <- DESeqDataSet(seB, design = ~ Donor + cell_type)
ddsB <- DESeq(ddsB)
resB <- lfcShrink(ddsB, coef = "cell_type_Plasmablast_vs_Bulk_B", type = "apeglm")

dfB <- as.data.frame(resB); dfB$PeakID <- rownames(dfB)
dfB <- dfB[!is.na(dfB$padj), ]
darsB <- dfB[dfB$padj < 0.05 & abs(dfB$log2FoldChange) >= 1, ]
cat("Design B DAR count:", nrow(darsB), "\n")

# --- 3. Compare the two designs ---
# If the unblocked design were detecting donor artefacts, controlling for donor
# would dissolve or reverse a substantial fraction of the calls.
darsA <- df[!is.na(df$padj) & df$padj < 0.05 & abs(df$log2FoldChange) >= 1, ]
ov <- intersect(rownames(darsA), rownames(darsB))
cat("\n=== COMPARISON ===\n")
cat("Design A (7 samples) DARs:", nrow(darsA), "\n")
cat("Design B (4 samples) DARs:", nrow(darsB), "\n")
cat("Shared DARs:", length(ov), "\n")
cat(sprintf("Design B DARs also found by A: %.1f%%\n", 100 * length(ov) / nrow(darsB)))

# Direction consistency across all of A's DARs that B could also test
common <- intersect(rownames(darsA), rownames(dfB))
agree  <- sign(darsA[common, "log2FoldChange"]) == sign(dfB[common, "log2FoldChange"])
cat(sprintf("log2FC sign agreement on A's DARs: %.1f%% (n=%d)\n",
            100 * mean(agree), length(common)))

write.table(darsB[order(darsB$padj), ], "DARs_designB_paired.tsv",
            sep = "\t", quote = FALSE, row.names = FALSE)
cat("\n>>> DARs_designB_paired.tsv saved\n")
```

A.8 Week 3 - annotation, target genes and motif enrichment

```
#!/bin/bash
# Week 3: peak-to-gene annotation, target gene extraction, HOMER motif enrichment

source /pcpool/home/sysbiol_tutor/miniconda3/bin/activate
conda activate /pcpool/home/sysbiol_share/envs/atac
mkdir -p ~/week3 && cd ~/week3
```

```
cp ~/week2/DARs_significant.tsv ~/week2/consensus_peaks.saf .

# --- Split DARs by direction, convert to BED ---
# DARs_significant.tsv columns: 1=baseMean 2=log2FoldChange 3=lfcSE 4=pvalue 5=padj 6=PeakID
# ($2+0) forces numeric comparison; scientific-notation values compare as strings otherwise
awk 'NR==FNR && FNR>1 {c[$1]=$2"\t"$3"\t"$4; next}
     FNR>1 && ($2+0)>=1 && $6 in c {print c[$6]"\t"$6"\t0\t+"}' \
     consensus_peaks.saf DARs_significant.tsv > DAR_up_plasmablast.bed

awk 'NR==FNR && FNR>1 {c[$1]=$2"\t"$3"\t"$4; next}
     FNR>1 && ($2+0)<=-1 && $6 in c {print c[$6]"\t"$6"\t0\t+"}' \
     consensus_peaks.saf DARs_significant.tsv > DAR_down_bulkB.bed

# Background for HOMER: the full consensus set
awk 'FNR>1 {print $2"\t"$3"\t"$4"\t"$1"\t0\t+"}' consensus_peaks.saf > consensus_all.bed

wc -l DAR_up_plasmablast.bed DAR_down_bulkB.bed consensus_all.bed  # 7976 / 6686 / 155948

# --- HOMER motif enrichment ---
# The shared environment has HOMER but no installed genome package, and
# configureHomer.pl cannot write to it. Pass the reference FASTA directly instead.
export HOMER_REF=/pcpool/home/sysbiol_tutor/Projects_Haubrock/homer_ref/Homo_sapiens.GRCh38.dna_sm.primary_asser

# -bg consensus_all.bed : compare changed regions against ALL open regions,
#                         not against random genomic background
# -size 200             : +/-100 bp around peak centre; wider windows dilute the motif
nohup findMotifsGenome.pl DAR_up_plasmablast.bed $HOMER_REF homer_up/ \
  -bg consensus_all.bed -size 200 -p 4 > homer_up.log 2>&1 &

nohup findMotifsGenome.pl DAR_down_bulkB.bed $HOMER_REF homer_down/ \
  -bg consensus_all.bed -size 200 -p 4 > homer_down.log 2>&1 &

# --- Gene annotation: build a TSS reference from the Ensembl GTF ---
wget -q https://ftp.ensembl.org/pub/release-110/gtf/homo_sapiens/Homo_sapiens.GRCh38.110.gtf.gz

# TSS depends on strand: gene start on '+', gene END on '-'.
# Getting this backwards would misplace every minus-strand TSS.
zcat Homo_sapiens.GRCh38.110.gtf.gz | awk -F'\t' '
  $3=="gene" {
    tss = ($7=="+") ? $4 : $5
    name=""; type=""
    if (match($9, /gene_name "[^"]+"/))    { name=substr($9,RSTART+11,RLENGTH-12) }
    if (match($9, /gene_biotype "[^"]+"/)) { type=substr($9,RSTART+14,RLENGTH-15) }
    if (name!="" && type=="protein_coding")
      print $1"\t"(tss-1)"\t"tss"\t"name"\t0\t"$7
  }' | sort -k1,1 -k2,2n > tss_protein_coding.bed          # 19,445 genes

# --- Assign each DAR to its nearest TSS ---
sort -k1,1 -k2,2n DAR_up_plasmablast.bed > DAR_up.sorted.bed
sort -k1,1 -k2,2n DAR_down_bulkB.bed     > DAR_down.sorted.bed

bedtools closest -a DAR_up.sorted.bed   -b tss_protein_coding.bed -d > DAR_up_annotated.txt
bedtools closest -a DAR_down.sorted.bed -b tss_protein_coding.bed -d > DAR_down_annotated.txt
# output: DAR cols 1-6, nearest TSS cols 7-12, distance col 13

# --- Gene lists for g:Profiler ---
# 5 kb cutoff: distal peaks may loop to a different gene entirely
awk '$13<=5000 {print $10}' DAR_up_annotated.txt   | sort -u > genes_up_promoter.txt
awk '$13<=5000 {print $10}' DAR_down_annotated.txt | sort -u > genes_down_promoter.txt
awk '{print $10}' DAR_up_annotated.txt   | sort -u > genes_up_all.txt
awk '{print $10}' DAR_down_annotated.txt | sort -u > genes_down_all.txt
```

```
# Remove bedtools' no-match placeholder
for f in genes_*.txt; do grep -v '^\.$' $f | grep -v '^$' > tmp && mv tmp $f; done

# --- Check which canonical markers are present, then choose targets ---
grep -wE "PRDM1|XBP1|IRF4|CD38|SDC1|MZB1|TNFRSF17" genes_up_promoter.txt
grep -wE "PAX5|BACH2|MS4A1|CD19|CR2|CIITA|EBF1"    genes_down_promoter.txt

cat > target_genes.txt <<'GENES'
PRDM1
IRF4
SDC1
PAX5
MS4A1
CIITA
GENES

# --- Extract every DAR near the six targets, with effect sizes ---
echo -e "gene\tdirection\tpeak\tchr\tstart\tend\tdist_to_TSS\tlog2FC\tpadj" > target_gene_DARs.tsv
for f in up down; do
  awk -v dir=$f 'NR==FNR {t[$1]; next} ($10 in t) {print $10"\t"dir"\t"$4"\t"$1"\t"$2"\t"$3"\t"$13}' \
    target_genes.txt DAR_${f}*_annotated.txt
done | sort > tmp_targets.txt

awk 'NR==FNR && FNR>1 {lfc[$6]=$2; p[$6]=$5; next}
     {print $0"\t"lfc[$3]"\t"p[$3]}' DARs_significant.tsv tmp_targets.txt >> target_gene_DARs.tsv

# --- Subset BAMs to target loci for IGV (full BAMs are 2-5 GB) ---
mkdir -p igv
export BAM_DIR=/pcpool/home/sysbiol_tutor/Projects_Haubrock/student_projects/sysbiol19/P04_Bulk_B_vs_Plasmablast
for s in SRR7650729 SRR7650735; do    # matched pair from donor 1001
  samtools view -b $BAM_DIR/${s}_REP1.mLb.clN.sorted.bam \
    6:105700000-106200000 11:60400000-60500000 9:36800000-37100000 \
    2:20100000-20250000 16:10800000-10950000 > igv/${s}.targets.bam
  samtools index igv/${s}.targets.bam
done
cp DAR_up_plasmablast.bed DAR_down_bulkB.bed igv/

# --- Extract top motif tables once HOMER finishes ---
head -16 homer_up/knownResults.txt   | cut -f1,3,5,7,9 > homer_up_top15.tsv
head -16 homer_down/knownResults.txt | cut -f1,3,5,7,9 > homer_down_top15.tsv
```

GO enrichment (g:Profiler, web interface). Three queries at https://biit.cs.ut.ee/gprofiler/gost, organism Homo sapiens:

1. `genes_up_promoter.txt` - 731 genes, 686 mapped
2. `genes_down_promoter.txt` - 733 genes, 696 mapped
3. The six target genes

Manhattan plots exported at 300 dpi; result tables exported as CSV with intersections.

Appendix B - Software Environments

```
# Week 1 and Week 2 Days 1-2: shared environment
source /pcpool/home/sysbiol_tutor/miniconda3/bin/activate
conda activate /pcpool/home/sysbiol_share/envs/atac
# cutadapt 4.7, bowtie2, samtools, bedtools, picard 3.4.0,
# subread/featureCounts 2.1.1, multiqc 1.35
# NOTE: macs2 2.2.7.1 in this environment is non-functional (see §7.2);
#       macs3 3.0.2 from /usr/bin was used instead.

# Week 2 Days 3-5: private R environment (DESeq2 not present in the shared env)
conda create -y -p ~/envs/deseq2 \
  -c conda-forge -c bioconda --override-channels \
```

```
  bioconductor-deseq2 bioconductor-apeglm bioconductor-vsn \
  bioconductor-enhancedvolcano r-pheatmap r-ggplot2 r-rcolorbrewer r-hexbin
conda activate ~/envs/deseq2
# R 4.5.3, DESeq2 1.50.2, apeglm 1.32.0, vsn 3.78.1, EnhancedVolcano 1.28.2
```

`--override-channels` restricts resolution to conda-forge and bioconda, which avoids the Anaconda Terms-of-Service prompt that blocked an earlier installation attempt.

```
# Week 3: back to the shared environment (HOMER, BEDTools, SAMtools)
conda activate /pcpool/home/sysbiol_share/envs/atac
# HOMER v4.11, BEDTools, SAMtools
# HOMER genome packages are not installed and configureHomer.pl cannot write
# to the shared directory, so the reference FASTA is passed directly:
#   /pcpool/home/sysbiol_tutor/Projects_Haubrock/homer_ref/Homo_sapiens.GRCh38.dna_sm.primary_assembly.fa
# Annotation: Ensembl GRCh38 release 110 GTF
# GO enrichment: g:Profiler web server (version e114_eg62_p19_27110d83)
# Genome browser: IGV with Human (GRCh38/hg38)
```

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